Probability & Statistics

Books on probability and statistics
https://www.quora.com/What-are-some-good-books-for-learning-probability-and-statistics

Introduction to Probability Theory

Probability Definitions
Probability of an event is the measure of the likelihood that the event will occur (in some experiment).

Probability is a number between 0 and 1.
But sometimes people express probability using percents.

0 – event is impossible
0.5 – the event is just as likely to occur as it is to not occur
1 – event will certainly happen, 100%.

The sum of probabilities of all outcomes of an experiment should always be 1.

Outcomes can be combinations of individual events.

For example, if we throw one dice cube – there are 6 possible outcomes (1,2,3,4,5,6), each one has same probability – 1/6.

If we throw 2 cubes, there are 6*6=36 possible combinations.
Probability of each combination is the same – 1/36.

If we define an outcome as a sum of numbers on top side of cubes after the throw, then there are 11 possible outcomes:

Sum	Combinations	Number of combinations	Probability
2	1+1	1	1/36
3	1+2,2+1	2	2/36
4	1+3,2+2,3+1	3	3/36
5	1+4,2+3,3+2,4+1	4	4/36
6	1+5,2+4,3+3,4+2,5+1	5	5/36
7	1+6,2+5,3+4,4+3,5+2,6+1	6	6/36
8	2+6,3+5,4+4,5+3,6+2	5	5/36
9	3+6,4+5,5+4,6+3	4	4/36
10	4+6,5+5,6+4	3	3/36
11	5+6,6+5	2	2/36
12	6+6	1	1/36
Total		36	1

From the table above you see how some outcomes can result from more combinations, thus have higher probability of happening.

Note: Events can be mutually exclusive, or not.
Also, events may be independent, or dependent on each other.

Marginal Probability = Simple Probability , a probability of an independent event occurring.

The term “marginal variable” is used to refer to a subset of variables in a table. They called “marginal” because they used to be calculated by summing values in a table along rows or columns , and writing the sum in the margins of the table.

Examples of assigning probability.

Example 1. Tossing a coin.

Possible outcomes : (tails or heads).
Events (tails of heads) are mutually exclusive.
Their probabilities should be equal.

$P_t + P_h = 1$
$P_t = 0.5$
$P_h = 0.5$

Example 2. Rolling a die.

Possible outcomes: (1,2,3,4,5,6).
They are mutually exclusive, probability equal, sum() = 1
So each probability = 1/6 = 0.167 3.

Example 3. Job Search.

I get 2 emails with job descriptions every day, 10 per week.

Two of them may be a fit: 2 of 10 (0.2).

So I send an email that I am interested.
I am invited to every third position I requested (0.2/3 = 0.067)

And I get 1 position out of 10 initial interviews (0.0067).

So the probability that an incoming email will result in a job is 0.0067, or ~1%.
So, I need to consider 100 positions (this will take 10 weeks), send 20 emails asking to invite, go to 6..7 interviews, and eventually get a job.

So job search takes 2..3 months.

Rules (Or, And, etc.)

Lets A be an event.
P(A) – probabiltiy of event A
then: P(A) + P(not A) = 1

If A and B are events, then:
P(A or B) = P(A) + P(B) – P(A and B)

for mutually exclusive events P(A and B) = 0,
so P(A or B) = P(A) + P(B)

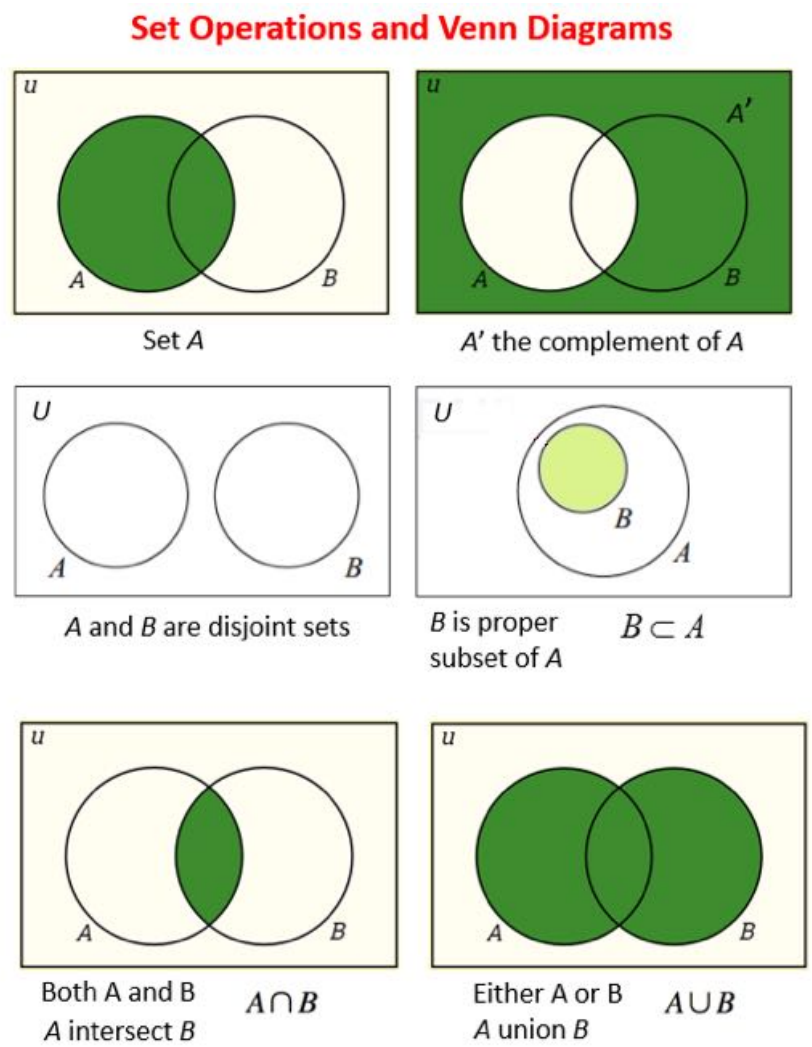
Venn Diagrams

Venn Diagrams = Graphical representation of events as circles.
See many examples – just google images for: Venn Diagram A or B.

For example, this image shows Intersection (A and B),
negation of intersection (A but not B, or B but not A),
union (A or B), and subtraction (A but not B) :

Conditional Probability
P(A|B) – probability of A given that B has occurred.
P(A|B) = P(A and B) / P(B) (if P(B) is not zero)

If A & B are independent form each other, then:
P(A|B) = P(A)
P(B|A) = P(B)



Bayes' Theorem (Law, Rule)
– https://en.wikipedia.org/wiki/Thomas_Bayes -
– https://en.wikipedia.org/wiki/Bayes%27_theorem –
– https://en.wikipedia.org/wiki/Bayesian_inference –

P(A|B) = P(B|A)*P(A)/P(B)

where A and B are events and P(B) not equal 0.

Frequentist interpretation of Bayes' theorem.
Probability measures a proportion of outcomes.
Suppose an experiment is performed many times.

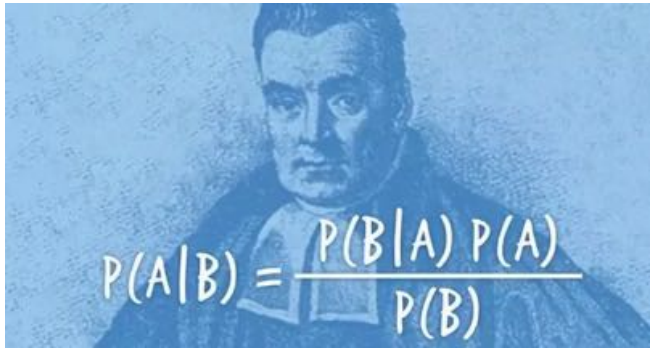
P(A) is the proportion of outcomes with property A,
and P(B) that with property B.

P(B|A) is the proportion of outcomes with property B
out of outcomes with property A,

and P(A|B) the proportion of those with A out of those with B.

Bayesian interpretation of Bayes' theorem.
Probability measures a degree of belief.
Bayes' theorem then links the degree of belief in a proposition before (A)
and after (B) accounting for evidence.

For example, suppose it is believed with 50% certainty
that a coin is twice as likely to land heads than tails.
If the coin is flipped a number of times and the outcomes observed,
then that degree of belief may rise, fall or
remain the same depending on the results.



Thomas_Bayes – (/'be?z/; 1701–1761)

For proposition A and evidence B,
P(A), the prior, is the initial degree of belief in A.
P(A|B), the posterior, is the degree of belief having accounted for B.
P(B|A)/P(B) – represents the support B provides for A.

See also Bayesian inference.
– https://en.wikipedia.org/wiki/Bayesian_inference

Bayesian inference is a method of statistical inference in which
Bayes' theorem is used to update the probability for a hypothesis
as more evidence or information becomes available.

Bayesian inference - calculate probability following chains of dependencies or correlations

Bayes' Theorem (conditional probability)

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

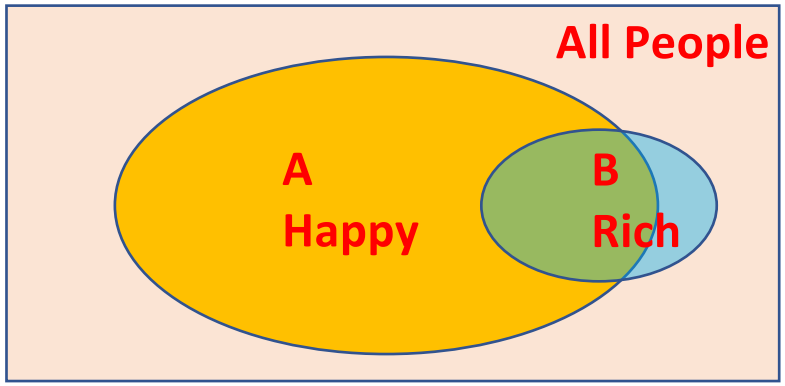
or

$$P(A|B) * P(B) = P(B|A) * P(A)$$

where A and B are events, and P(B) not equal 0.

$$\text{posterior} = \frac{\text{prior} \times \text{likelihood}}{\text{evidence}}$$

Thomas Bayes
(1701–1761)



Intuitive explanation of Bayesian rule (on the left).
A = Happy, B = Rich
P(A) = 40% = probability of being happy (Yellow)
P(B) = 5% = probability of being rich (Blue)
P(A & B) = 4% = probability of rich & happy (Green)
P(B|A) = 10% = probability of being rich given that Happy
P(A|B) = 80% = probability of being happy given that rich

You can calculate green intersection using two methods:
P(A|B)*P(B) = 80%*5% = 4%
P(B|A)*P(A) = 10%*40% = 4%

Probability - Bayesian vs Frequentist

The **Frequentist** see probability as something that has to do with a limiting frequency based on an observed proportion.

The **Bayesian** is subjective and uses a priori beliefs to define a prior probability distribution on the possible values of the unknown parameters.

Frequentists accuse Bayesians for being subjective in selecting a prior model / distribution.

Bayesians snap back that at least they have a model, whereas Frequentists just calculate averages of observed data

Allen Downey – https://www.youtube.com/watch?v=R6d-AbkhBQ8

Frequentist (classical):

Calculate means (estimates) and confidence intervals
Problems treating missing data (**survivorship bias**)
Problems with hypothesis testing (**p-hacking**)

Bayesian:

Enumerate all possible hypotheses – and compute their probabilities.
We make a (subjective) choice of "prior" (model, distribution) and produce posterior distribution.
Allows for complex decision analysis under uncertainty.

Example1: Polling, two candidates: Alice & Bob.
52% of polled voters intend to vote for Alice.
Confidence interval [42%,62%], p-value = 0.34
What is the probability that Alice will win ?
Classical frequentist approach can not answer this question.
Bayesian method – result is a posterior distribution of possible outcomes based on data.
Thus you can get your answer.

Example2: comparing two drugs A & B.
A is better than B with p value < 0.001
A is more expensive.
Which drug should we prescribe?
Cost-benefit.
Suppose we get predicted probability of survival with A and B.
We could then compute dollars per life saved.

Example3: Guessing game
You see the prize – and guess the price
Opponent sees a different prize – and guesses the price.
Whoever is closer (without going over) wins.

Example4: MCMC (Markov-Chain Monte-Carlo)

Bayesian vs Frequentist – Survivorship Bias in World War 2

During World War II the US military's concluded that the most-hit areas of the plane needed additional armor.
But statistician Abraham Wald (Statistical Research Group (SRG) at Columbia University) has recommended adding armor to the areas that showed the least damage. His reasoning was that the sample didn't include the planes which were shot down.

It is dangerous to use data "as is" to create a model,
because the data may be biased to start with,
and it can easily lead to creating a biased model

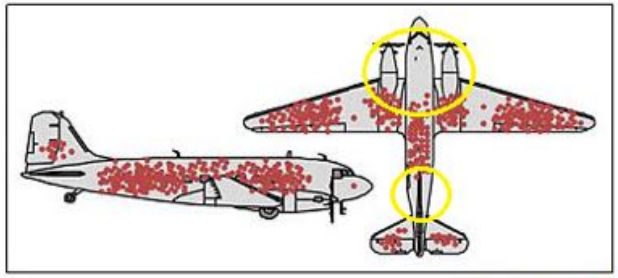
Problem occur when we have:
- under-represented features (missing data)
- over-represented features,
- "target leakage" effect.

Survivorship bias:
- https://www.youtube.com/watch?v=B3YQJ5DwTzM

Gender Bias:
- https://www.youtube.com/watch?v=p60-Jdq754A

Bias in criminal justice:
- https://www.youtube.com/watch?v=Gi4YeRqfb24

Target Leakage:
- https://www.youtube.com/watch?v=y8qaI5mpJeA



Credit: Cameron Moll

Gentlemen, you need to put more armour-plate where the holes aren't because that's where the holes were on the airplanes that didn't return - Abraham Wald 1942.

Random Variable. Probability Distribution

Random variable is a numerical variable taking different values. Values may be **discrete or continuous**.

For example, think of variable “x” which may have any real value between -10 and +10.

Or think of a variable “z” which can have discrete values 0,0.1,0.2, … 0.9, 1.0

Discrete Case

Probability function P(x) of a discrete random variable “x” is simply a function which gives probability of discrete values.

For example, if the random variable “x” may have two values: (0 or 1), and if probabilities of those values are equal, then

P(0) = 0.5, f(1) = 0.5.

The sum of individual probabilities for all possible values should be 1.

Expected value is an weighted average of all values.

E(x) = sum(Xi*P(Xi)), where Xi goes through all allowed values of X.

Variance = sigma squared = sum((Xi-E(x))**2 * P(Xi))

Standard Deviation = sqrt(sigma squared)

Probability Density Function (PDF)
vs. Cumulative Probability Function (CPF)

Continuous Case

Probability Density Function (PDF) f(x) is a continuous function.

f(x)*dx gives probability that value is between x and x+dx.

The total integral of f(x)*dx should be equal to 1.

Cumulative Probability Function (CPF)
This is simply a function F(x) which gives probability that value is less than x. It monotonously grows from 0 to 1 as x grows (from small to large values).

The PDF is a first derivative of CPF.

CPF(x+dx) – CPF(x) = = [Prob. of value < (x+dx)] – [Prob. of value < x] = f(x)*dx

Mean value = Expected value

For discrete case we had:
E(x) = sum(Xi*P(Xi))

For continuous case this translates into
E(x) = Integral(xF(x)dx)

Binomial Distribution

– https://en.wikipedia.org/wiki/Binomial_distribution

if:
Experiment consists of N trials.
Each trial can have two outcomes (True of False).
The probabilities of these outcomes are the same for all trials (p and q = 1-p).
The trials are independent from each other.

Then:
The probability of having "x" True values (out of N) is:

f(x) = C(N,x)*p^x*(1-p)^(N-x)

where C(N,x) = N!/(x!*(N-x)!) – a binomial coefficient

At large numbers the distribution f(x) has a bell-curve shape

Mean value = Expected value E(x) = N*p
Variance = N*p*(1-p).

Combinatorics

Factorial

N! = N*(N-1)*(N-2)*…*1

Reordering

Let's say, we have N objects (N-element unordered set).

Question: In how many ways can we arrange them?
Answer: N!

Explanation:
N variants for the 1st element (N-1) variants for the 2nd element etc.
Example: How many ways are there to arrange the letters ABCDE in a row?

Answer: 5!=120

Subsets

Question: In how many ways can we select m-elements out of N-element set?

Answer: C(N,m) = N!/(m!*(N-m)!) – a binomial coefficient

Explanation:
N variants for the first element of subset
(N-1) for the 2nd etc.
(N-m+1) for the m-th element
The above can be written as N!/(N-m)!
Then we need to divide by the number of reordering permutations
within the subset, that is by m!

Poisson and Exponential Distributions
– https://en.wikipedia.org/wiki/Poisson_distribution
– https://en.wikipedia.org/wiki/Exponential_distribution

Suppose you are sitting near the road.
You notice that on average you see 10 cars passing by every minute
(we say: the rate lambda = 10/min).
Knowing the average rate, what would be the probability
of having only 5 cars in one minute?
Or 15 cars? Or any given number?

Poisson Probability P(k) is the probability of a given
number of events occurring in a fixed interval (of time)
if these events occur with a known average rate
and independently of each other
and of the time since the previous events.

Poisson Formula

P(x) = (lambda**x)*exp(-lambda)/x!

For example, for x=0 (probability of zero events in given interval):

P(0) = (lambda**0)*exp(-lambda)/0! = 1*exp(-lambda)/1 = exp(-lambda)

Poisson Distribution

Intervals between Poisson events distributed exponentially.

If we have events happening which follow Poisson Distribution,
we can measure the time intervals between these events.

Question: How these times will be distributed?

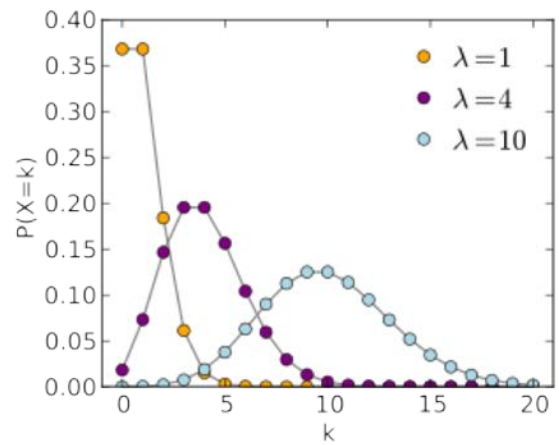
Answer: These times will have exponential distribution
with Probability Density Function (PDF) as:

f(t) = lambda*exp(-lambda*t)

The average time interval will be (1/lambda).
Standard deviation happens to be the same (1/lambda).

$$\lambda = \frac{\sum_{i=1}^{n} x_i}{n}$$

$$f(x;\lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$$



Example – The Matching Problem

https://www.randomservices.org/random/urn/Matching.html

An absent-minded secretary prepares N letters and N addressed envelopes, and then randomly stuffs the letters into the envelopes.
A match occurs if a letter is inserted in the proper envelope.

What is the probability to have "k" matches ?
(0<= k <= N)

$$\mathbb{P}(N_n = k) \to \frac{e^{-1}}{k!} \text{ as } n \to \infty, \quad k \in \mathbb{N}$$

Example distribution for N = 5:

0	1	2	3	4	5
44	45	20	10	0	1

Derangement - a permutation of the elements of a set, such that **no element appears in its original position**.
https://en.wikipedia.org/wiki/Derangement

Calculating number of derangements **!n**
(where (1/e) comes from) :

$$n!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} \cdots \pm \frac{1}{n!}\right)$$

The Number of Derangements of an *n*-Element Set (sequence A000166 in the OEIS) for Small *n*

n	0	1	2	3	4	5	6	7	8	9	10	11	12	13
!n	1	0	1	2	9	44	265	1,854	14,833	133,496	1,334,961	14,684,570	176,214,841	2,290,792,932

Binomial Distribution => Gauss

$$f(x) = C(n,x)p^x q^{n-x} \sim \frac{1}{\sqrt{2\pi npq}} e^{-(x-np)^2/2npq}$$

using Stirling's formula:

$$n! = n^n e^{-n}\sqrt{2\pi n}\left[1 + \mathcal{O}\left(\frac{1}{n}\right)\right]$$

Poisson Distribution => Gauss

The Poisson Distribution can be approximated with Normal when λ is large (greater than 10).

$$P_n = \frac{\lambda^n e^{-\lambda}}{n!}$$

x = n = λ(1 + δ) , where λ >> 1 and δ << 1
Using Stirling's formula for n!

$$\begin{aligned} p(x) &= \frac{\lambda^{\lambda(1+\delta)} e^{-\lambda}}{\sqrt{2\pi} e^{-\lambda(1+\delta)} [\lambda(1+\delta)]^{\lambda(1+\delta)+1/2}} \\ &= \frac{e^{\lambda\delta}(1+\delta)^{-\lambda(1+\delta)-1/2}}{\sqrt{2\pi\lambda}} = \frac{e^{-\lambda\delta^2/2}}{\sqrt{2\pi\lambda}} = \frac{e^{-(x-\lambda)^2/(2\lambda)}}{\sqrt{2\pi\lambda}} \end{aligned}$$

Binomial Distribution => Poisson

The Poisson distribution is a limiting case of the binomial distribution which arises when the number of trials n increases indefinitely whilst the product **μ = np**, which is the expected value of the number of successes from the trials, remains constant.

$$b(x;n,p) = \frac{n!}{(n-x)!x!} p^x (1-p)^{n-x}$$

μ = np = const, n --> inf

$$\underbrace{\frac{\mu^x}{x!}\frac{n!}{(n-x)!n^x}}_{1}\underbrace{\left(1-\frac{\mu}{n}\right)^n}_{e^{-\mu}}\underbrace{\left(1-\frac{\mu}{n}\right)^{-x}}_{1} \Rightarrow \underbrace{\frac{\mu^x e^{-\mu}}{x!}}_{\text{Poisson}}$$

Coin tossing

We are tossing a fair coin: tail = 0, head = 1
p = (1-p) = 0.5
Expected μ = p*N = N/2
For large N we approximate with Gauss
Variance σ^2 = Npq = N/4
Standard error σ = √ N / 2
Relative error σ / μ = (√ N / 2) / (N/2) = 1/ √ N
So If we want error ~1%, we need to do N ~ 10,000

Also for χ^2 calculations:
$(x - \mu)^2 / \mu$ --> σ^2 / μ = (N/4) / (N/2) = ½ = const

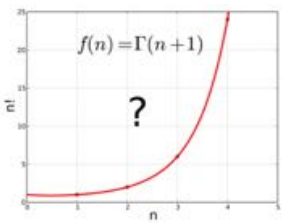
Gamma Function & Distribution

Gamma function – an extension of a **factorial**.
It is a smooth curve that connects the points (x, y) given by y = (x − 1)! at the positive integer values for x.

$$\Gamma(n) = (n-1)!$$

Derived by Daniel Bernoulli and Leonhard Euler.
For complex numbers with a positive real part the gamma function is defined via the integral.

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x}\, dx, \qquad \Re(z) > 0$$

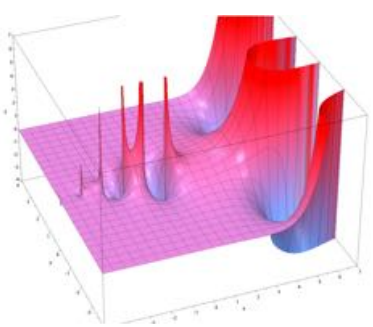


Daniel Bernoulli
(1700 - 1782)

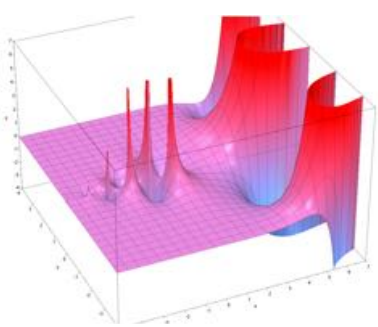
Leonhard Euler
(1707 - 1783)

The gamma function then is defined as the analytic continuation of this integral function.

Real Part of **Γ**



Imaginary Part of **Γ**



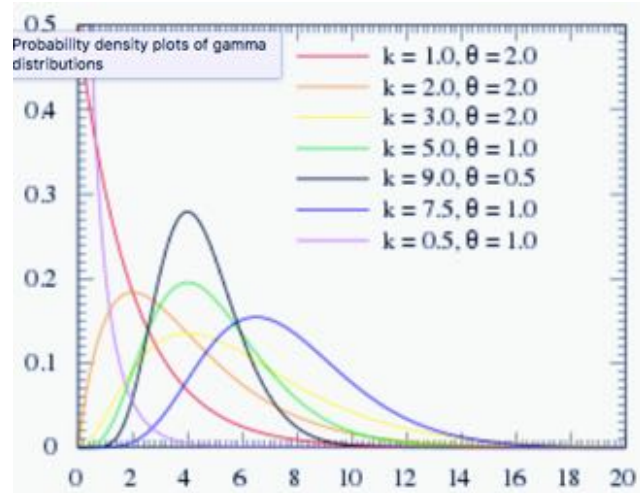
Gamma distribution: time required for **"k"** events to occur in a Poisson process with mean arrival time of **"θ"**. Example: if cars are passing by every 10 seconds, how long will you wait on average to count 10 cars?

$x \in (0, \infty)$	$x \in (0, \infty)$
$f(x) = \frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-\frac{x}{\theta}}$	$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$

k>0 : shape parameter
θ>0 : scale parameter
kθ = mean value
$k\theta^2$ = variance

$\alpha = k$: shape parameter
$\beta = 1/\theta$: rate parameter
α / β = mean value
α / β^2 = variance

The exponential, Erlang (k=positive in), and chi-squared distributions are special cases of the gamma distribution.
k = 1 : exponential distribution
k > 1 : unimodal with mode at (k-1)* θ
k →∞ , the gamma distribution approaches a normal distribution in shape.

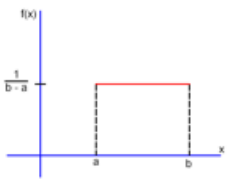


Uniform and Normal Distributions.

Uniform Distribution

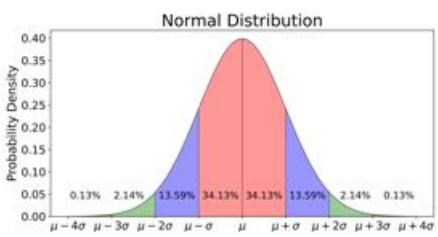
If some value can change between some minimum and maximum values a & b, and the probability of any value in interval [a,b] is the same, then we say that we have uniform distribution.

The PDF (Probability Density Function) for Uniform Distribution looks like a rectangle with the height (1/(b-a)), so that the total area under the curve is exactly 1.



Normal (Gaussian Bell-shaped) Distribution

The basic shape of the PDF is exp(-x*x/2)
You need to add a coefficient to normalize it to 1
(means – make area under the curve equal to exactly 1).
You can also have the curves have different position
(average value) and different width (sigma).



Central Limit Theorem (CLT).

The normal distribution is very common in nature.
Because when we have big number of sets of variables,
the distribution of parameters of these sets (for example, averages)
tend to converge to this bell form.
This fact is called **Central Limit Theorem (CLT)**.

Here is a simple example of CLT:

Suppose we have N1 sets, each set has N2 values, all values drawn from the same distribution. The shape of distribution may be anything (uniform, bell-curve, triangular, exponential, etc.).

We calculate an average in each set – so we have N1 average values.
Question: How will these average values be distributed?

Answer: As N1 & N2 become big enough (30+), the distribution will get close to the Normal Gaussian bell curve.

There are some nice animated demonstrations of CLT - just google for them.

De Moivre–Laplace theorem

De Moire-Laplace theorem is a simple demonstration how a symmetrical binomial distribution can be approximated by Gaussian bell curve (under certain conditions).
See video demonstration below.

https://en.wikipedia.org/wiki/De_Moivre–Laplace_theorem

Introduction into Statistics

"Statistics is the study of the collection, analysis, interpretation, presentation, and organization of data."

Being a Statistician is the major part of being a Data Scientist.
And it is tricky, because there is easy to misuse statistics:
"There are three kinds of lies: lies, damned lies, and statistics."

Statistics involves

- Collecting (sampling) the data from different kind of observations
- Visualizing the data, estimating the parameters (average/mean values, errors, etc.)
- Testing validity of a hypothesis (null hypothesis) – or alternative hypothesis
- Estimating intervals, estimating significance of observed differences, etc.

Here are some basic formulas:

formula to calculate mean

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

formula to estimate sample standard deviation

$$\sigma = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2}$$

formula to calculate sample SD variance

$$\sigma^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2$$

Intuitive explanation for dividing by (n-1)
when calculating standard deviation:

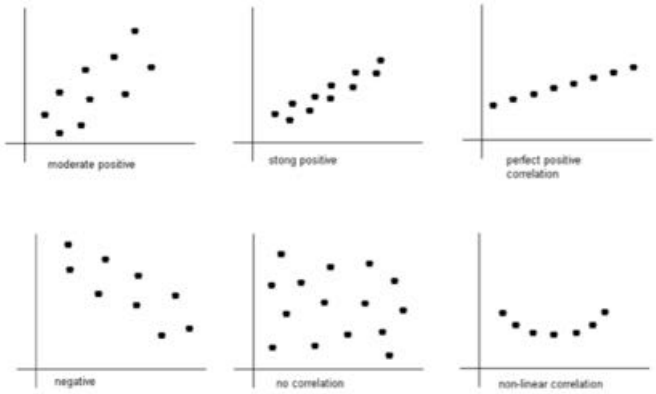
– http://stats.stackexchange.com/questions/3931/intuitive-explanation-for-dividing-by-n-1-when-calculating-standard-deviation

Correlation and Regression

Pearson Correlation **r**

https://en.wikipedia.org/wiki/Pearson_correlation_coefficient

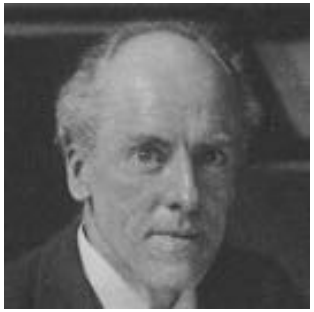
$$r = \frac{SP}{\sqrt{SS_X SS_Y}} = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2 \sum(y-\bar{y})^2}}$$



$$\rho_{X,Y} = \frac{cov(X,Y)}{\sigma_X \sigma_Y} = \frac{E[(X-\mu_X)(Y-\mu_Y)]}{\sigma_X \sigma_Y}$$

Pearson correlation is also known under names:

- Pearson correlation coefficient (PCC)
- Pearson's r
- Pearson product-moment correlation coefficient (PPMCC)
- Bivariate correlation



Karl Pearson
(1857 - 1936)
Famous English mathematician and biostatistician

r can vary from -1 to 1:
-1 – perfect negative correlation
+1 – perfect positive correlation

R^2 is the square of the Pearson's correlation coefficient **r** (for a single regressor, fitted by least squares)

Linear Regression

Linear Regression = draw a line through experimental points.

- https://en.wikipedia.org/wiki/Linear_regression
- https://en.wikipedia.org/wiki/Simple_linear_regression
- https://paternogbc.shinyapps.io/SS_regression/ - play with it

We derive linear model by minimization the mean square error (MSE):

$$\min_{\hat{\beta}_0,\hat{\beta}_1} \sum_{i=1}^{N} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2.$$

We take partial derivatives by two coefficients - and solve system of 2 equations, and after some algebraic transformations, we derive the standard **OLS** method (**OLS = Ordinary Least Squares**).

y_m = y_mean = sum(y)/N
x_m = x_mean = sum(x)/N

var(x) = variance(x) = sum($(x-x_m)^2$)
covar(x,y) = covariance(x,y) = sum($(y-y_m)*(x-x_m)$)

slope b_1 = covar(x,y) / var(x)
intercept $b_0 = y_m - b_1 * x_m$

$$b_1 = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sum (x-\bar{x})^2}$$

Standard Error of regression slope:

$$s(b_1) = \sqrt{\frac{1}{n-2} \cdot \frac{\sum (y_i - \hat{y}_i)^2}{\sum (x_i - \bar{x})^2}} \quad = \quad \sigma_{b_1} = \frac{\sigma_\varepsilon}{\sqrt{\sum (x-\bar{x})^2}}$$

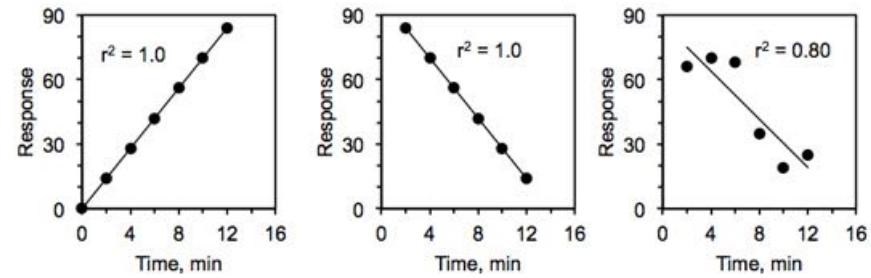
where sd(...) is standard deviation, and r – correlation coefficient

When drawing a curve through experimental points, it is very important to estimate how good this curve fits the data, and if fitting the data with this curve makes sense. For this you can use the **Coefficient of Determination R^2**:

– https://en.wikipedia.org/wiki/Coefficient_of_determination

"**R squared**" is a number between 0 and 1.
R^2 close to 1 – good model.

R^2 = **r^2** (the square of the Pearson's correlation coefficient **r** (for a single regressor, fitted by least squares)



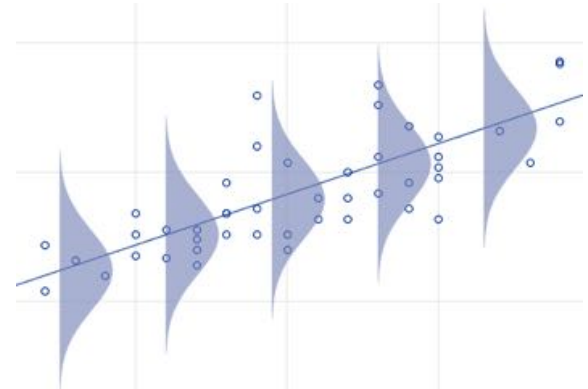
$$r^2 = \frac{\text{SSM}}{\text{SST}} = \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum (y_i - \bar{y})^2}$$

$$R^2 = 1 - \frac{SS_{RES}}{SS_{TOT}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

where:
SS = Sum of Squared
RES – residual errors = diffs between model and points)
TOT – total, diffs between mean and points
y_bar – mean
y_hat – model

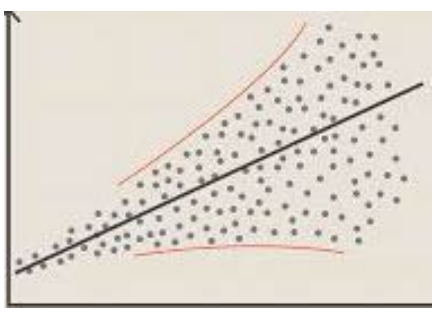
Linear Regression - assumptions

- **Linearity**: The relationship between X and the mean of Y is linear.
- **Normality**: For any fixed value of X, Y is normally distributed. We presume that deviations ϵ_i are picked from normal distribution. From this follows that the sampling distributions of slope and intercept (b_1 & b_0) are also normally distributed.
  Note: even if the ϵ_i are not Normally distributed, a general form of the central limit theorem tells us that the distributions of slope and intercept will be approximately normal if number "n" is big enough.
- **Independence**: Observations are independent of each other.
  We assume that our sample is a SRS (simple random sample) of size "n". This means that it consists of "n" points chosen from the population in such a way that every set of n individuals has an equal chance to be the sample actually selected.
- **Homoscedasticity**: The variance of residual is the same for any value of X.

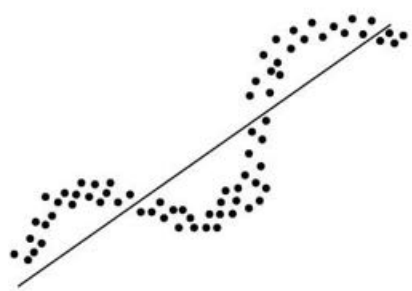


Linearity, normality,
same deviation (same σ) - **Homoscedasticity**

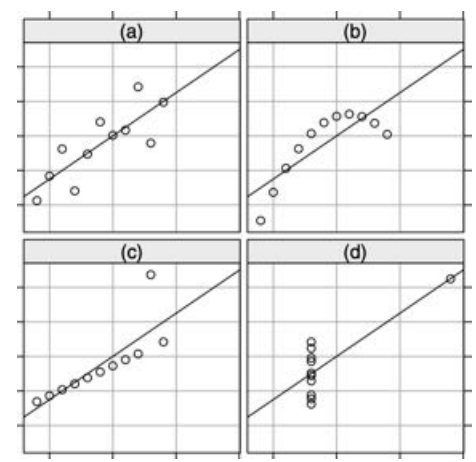
Heteroscedasticity
Variance changes



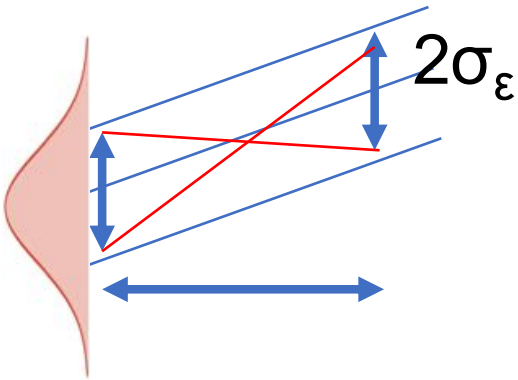
Autocorrelation.
Observations are NOT independent



Four sets of points – fit by the same line.



Linear Regression – Slope Error



ϵ_i are Normally distributed with standard error σ. "x" is distributed uniformly ?
The error of the slope is calculated as:

$$\sqrt{\frac{\frac{1}{n-2}\sum_{i=1}^{n}\hat{\varepsilon}_i^{\,2}}{\sum_{i=1}^{n}(x_i-\bar{x})^2}} = \frac{\sigma_\varepsilon}{\sqrt{\sum(x-\bar{x})^2}}$$

Note:
even if the ϵ_i are not Normally distributed, a general form of the central limit theorem tells us that the distributions of slope and intercept will be approximately Normal if number "n" is big enough.

We do not know real population **σ**, we use the estimated model standard deviation **s**.
When we do this, we move from the Normal distribution to **t**-distributions with degrees of freedom **n – 2**, the degrees of freedom of **s**.

$$s = \sqrt{\frac{\sum e_i^2}{n-2}}$$

A confidence interval for the slope β1 is b1 ± t*SEb1
In this expression, t is the value for the t-distribution density curve with (n – 2) degrees of freedom with area between [−t, t].

To test the hypothesis H0: β1 = 0, compute the test statistic :

$$t = \frac{b_1}{\mathrm{SE}_{b_1}}$$

Example: 42 points, df= n-2 = 40.
Looking for a critical value in t-test table for df=40, P=0.05, two-tails combined: cv = 2.021

Then if slope b1=35, error SEb1=10,
then t = 35/10 = 3.5 >> 2.021, so we reject the null hypothesis that slope = 0

STANDARD DEVIATION OF THE REGRESSION SLOPE COEFFICIENT (POPULATION)

$$\sigma_{b_1} = \frac{\sigma_\varepsilon}{\sqrt{\sum(x-\bar{x})^2}}$$

where:

σ_{b_1}= Standard deviation of the regression slope (Called the standard error of the slope)

σ_ε= Population standard error of the estimate

ESTIMATOR FOR THE STANDARD DEVIATION OF THE REGRESSION SLOPE

$$s_{b_1} = \frac{s_\varepsilon}{\sqrt{\sum(x-\bar{x})^2}} = \frac{s_\varepsilon}{\sqrt{\sum x^2 - \frac{(\sum x)^2}{n}}}$$

where:

s_{b_1}= Estimate of the standard error of the least squares slope

s_ε= $\sqrt{\frac{SSE}{n-2}}$ Sample standard error of the estimate

Linear Regression – Slope Error via "r"

Standard Error of regression slope:

$$s(b_1) = \sqrt{\frac{1}{n-2} \cdot \frac{\sum (y_i - \hat{y}_i)^2}{\sum (x_i - \bar{x})^2}} \quad = \ldots = \sqrt{\frac{1}{n-2} \frac{sd(y)}{sd(x)} \sqrt{1 - r^2}}$$

where sd(...) is standard deviation, and **r – correlation coefficient.**

https://stats.stackexchange.com/questions/342632/how-to-understand-se-of-regression-slope-equation/342672

https://www.slideserve.com/karis/introduction-to-linear-regression-and-correlation-analysis

$$s(b_1) = \sqrt{\frac{1}{n-2} \cdot \frac{\sum (y_i - \bar{y})^2}{\sum (x_i - \bar{x})^2} \cdot \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}}$$

$$= \sqrt{\frac{1}{n-2}} \sqrt{\frac{\frac{1}{n-1} \sum (y_i - \bar{y})^2}{\frac{1}{n-1} \sum (x_i - \bar{x})^2}} \sqrt{\frac{SS_{\text{error}}}{SS_{\text{total}}}}$$

$$= \sqrt{\frac{1}{n-2}} \sqrt{\frac{sd(y)^2}{sd(x)^2}} \sqrt{\frac{SS_{\text{total}} - SS_{\text{model}}}{SS_{\text{total}}}}$$

$$= \sqrt{\frac{1}{n-2}} \frac{sd(y)}{sd(x)} \sqrt{1 - r^2}$$

TEST STATISTIC FOR CORRELATION

$$t = \frac{r}{\sqrt{\frac{1 - r^2}{n-2}}}$$

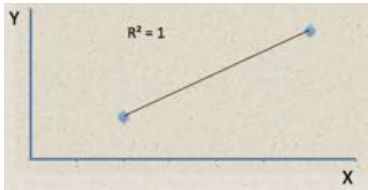
$$df = n - 2$$

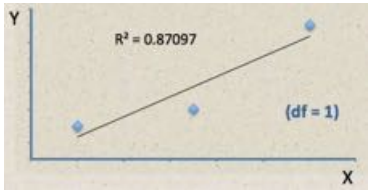
where:

t = Number of standard deviations r is from 0
r = Simple correlation coefficient
n = Sample size

df = degrees of freedom = n-2

If we have 2 points – we can draw only one line, no freedom, df = 0.
As we add points – we get more freedom.

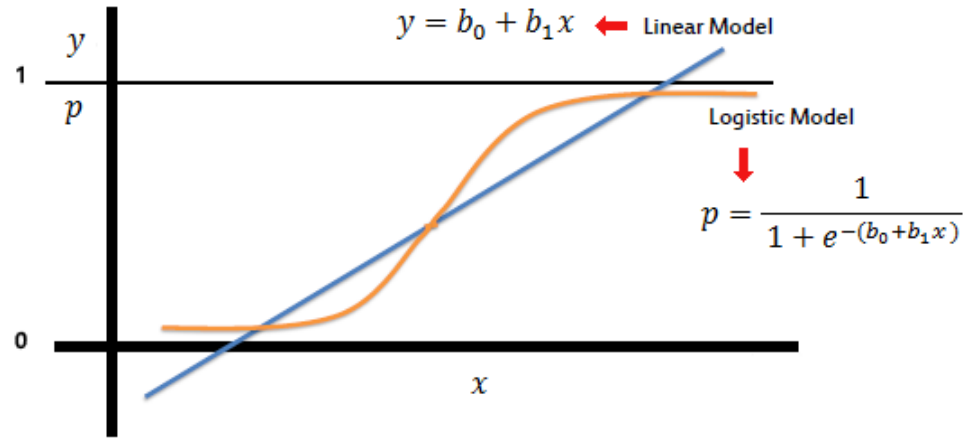




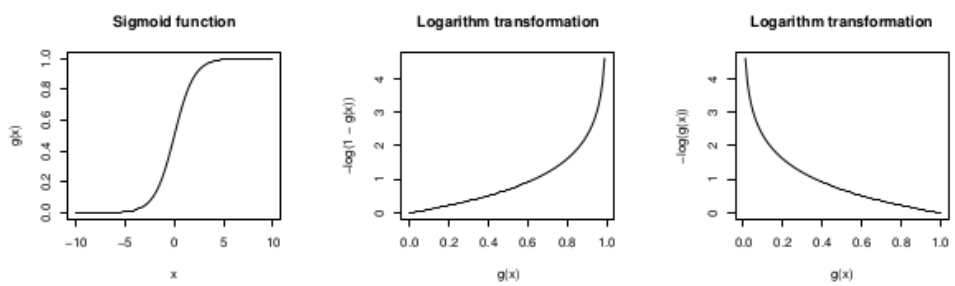
Logistic Regression
(binary classification model)

$$ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 x$$

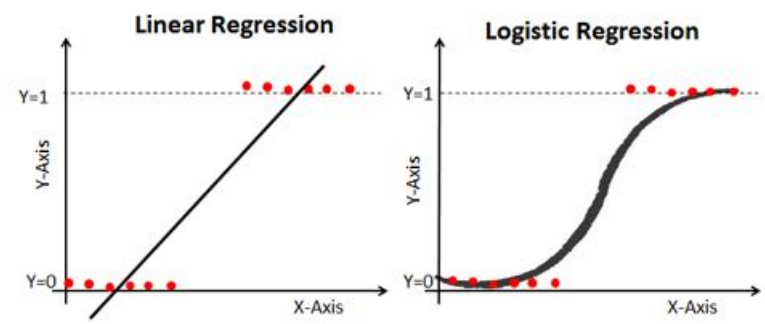
$$=> P = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$



Coefficients "beta" are fitted using gradient descent.
Common error function consists of two separate functions
(for target values 0 and 1):



(a) *Sigmoid function.* (b) *Cost for $y = 0$.* (c) *Cost for $y = 1$.*

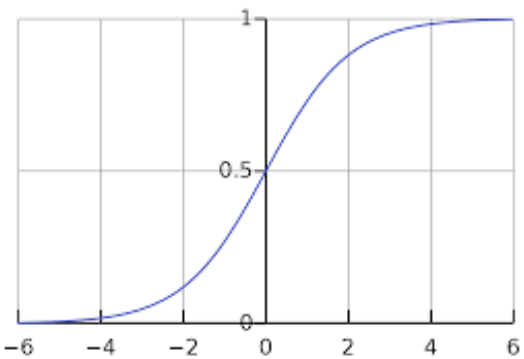


Standard Error (SE) for probability p and coefficient b_1

$SE(p) = SE(b_1) * (p*(1-p))$

Logistic Function

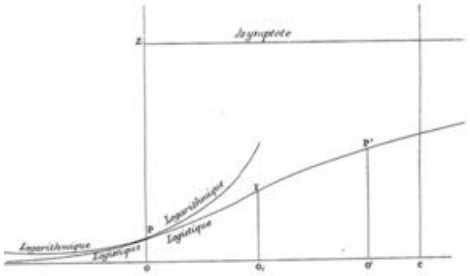
The logistic function was introduced in a series of three papers by **Pierre François Verhulst** between 1838 and 1847, who devised it as a model of population growth by adjusting the exponential growth model, under the guidance of **Adolphe Quetelet**



$$Q(t) = \frac{1}{1 + e^{-t}}$$

Pierre François Verhulst
1804-1849, Brussels,
Belgian mathematician

Lambert Adolphe Jacques
Quetelet
1796-1874
Belgian astronomer,
mathematician, statistician
and sociologist



Graph from original work by Verhulst
introducing "Logistique" function

Logistic Regression – Assumptions

Logistic regression does NOT make many of the key assumptions
of linear regression and general linear models
that are based on ordinary least squares algorithms.

- First, logistic regression does not require a linear relationship between the dependent and independent variables.
- Second, the error terms (residuals) do not need to be normally distributed.
- Third, homoscedasticity is not required.
- Finally, the dependent variable in logistic regression is not measured on an interval or ratio scale.

However, some other assumptions still apply.

- First, binary logistic regression requires the dependent variable to be binary
  and ordinal logistic regression requires the dependent variable to be ordinal.

- Second, logistic regression requires the observations to be independent of each other.
  In other words, the observations should not come from repeated measurements or matched data.

- Third, logistic regression requires there to be little or no **multicollinearity** among the independent variables. This means that the independent variables **should not be too highly correlated with each other**. If we have two highly variables x1 and x2, then the model may have very high variance with their coefficients (for example b1 & -b2, 10b1, -10b2, 1000b1, -1000b2, etc.).

- Fourth, logistic regression assumes linearity of independent variables and log odds.

- Finally, logistic regression typically requires a large sample size. A general guideline is that you need at minimum of 10 cases with the least frequent outcome for each independent variable in your model. For example, if you have 5 independent variables and the expected probability of your least frequent outcome is .10, then you would need a minimum sample size of 500 (10*5 / .10).

Logistic Regression – finding coefficients

There is no analytical solution.

The regression coefficients are usually estimated
using **Maximum Likelihood Estimation (MLE)**.

Unlike **linear regression** with normally distributed residuals,
it is not possible to find a closed-form expression for the coefficient values
that maximize the likelihood function,
so that an **iterative process must be used** instead; for example Newton's method.

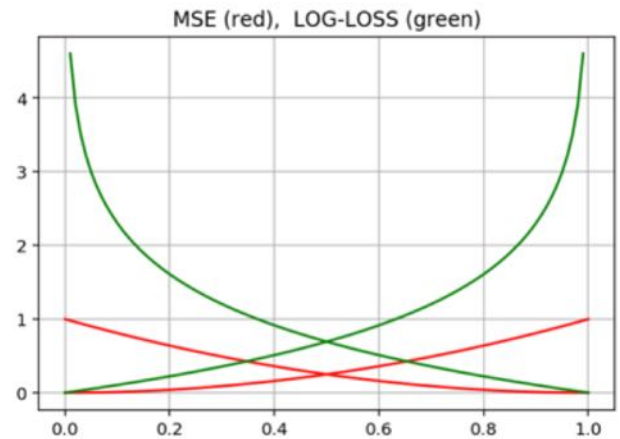
The model may not converge in some cases:

- not enough data (small number of rows for given number of predictor variables)
- multicollinearity (some variables are strongly correlated with each other)
- sparseness (missing data)
- complete separation (predictors perfectly predict the criterion)

In machine learning applications where logistic regression
is used for binary classification,
the MLE minimizes the **Cross Entropy Loss** function.

$$L(\mathbf{w}) \;=\; \frac{1}{N}\sum_{n=1}^{N} H(p_n, q_n) \;=\; -\frac{1}{N}\sum_{n=1}^{N}\Big[y_n \log \hat{y}_n + (1-y_n)\log(1-\hat{y}_n)\Big]$$

Log-Loss error function is better than MSE:



If “q” is model prediction, and “p” is probability of actual label, then similarity between q & p can be expressed as **cross-entropy** H(p,q):

$$H(p,q) \;=\; -\sum_i p_i \log q_i \;=\; -\,y\log\hat{y} - (1-y)\log(1-\hat{y})$$

Logistic Regression - Z-value

- http://logisticregressionanalysis.com/1577-what-are-z-values-in-logistic-regression/

z-value is the regression coefficient divided by its standard error.
It is also sometimes called the z-statistic.
It is usually given in the third column of the logistic regression coefficient table output.

A good rule of thumb is to use a cut-off value of 2,
which approximately corresponds to a two-sided hypothesis test
with a significance level of alpha=0.05.

$$Z = \frac{\hat{\beta} - 0}{\hat{\sigma}_{\hat{\beta}}} = \hat{\beta} / \hat{\sigma}_{\hat{\beta}}$$

Here is a very good page to test logistic regression:
- https://statpages.info/logistic.html

This page contains a straightforward JavaScript implementation of a standard iterative method to maximize the Log Likelihood Function (LLF), defined as the sum of the logarithms of the predicted probabilities of occurrence for those cases where the event occurred and the logarithms of the predicted probabilities of non-occurrence for those cases where the event did not occur.

Maximization is by Newton's method, with a very simple elimination algorithm to invert and solve the simultaneous equations. **Central-limit estimates of parameter standard errors** are obtained **from the diagonal terms of the inverse matrix**. Odds Ratios and their confidence limits are obtained by exponentiating the parameters and their lower and upper confidence limits, approximated by +/- 1.96 standard errors (for 95% limits).

Example: Logistic Regression output

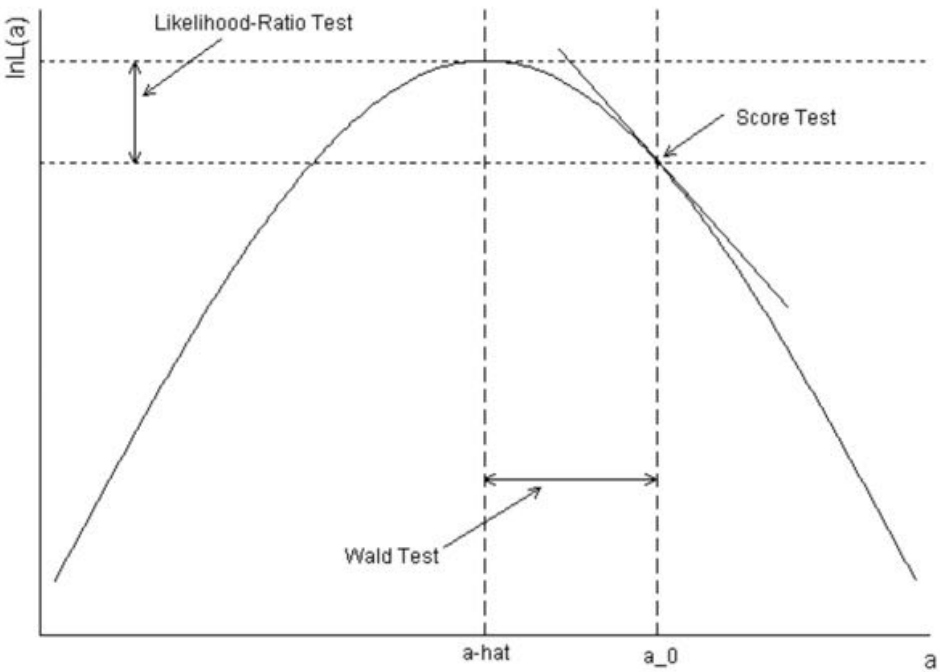
Columns:

estimates of coefficients,
std. error,
z-value,
p-value,
odds ratio - relative change of the odds when predictor increased by 1.0 units.

Coefficients:					
	Estimate	Std. Error	z value	Pr(>|z|)	Odds Ratio
(Intercept)	-17.91	2.223	-8.06	0	
Income	0.000202	0.000024	8.55	0	1.0002
IsFemale	1.646	0.4651	3.54	0.0004	5.1862
IsMarried	0.5662	0.5864	0.97	0.3343	1.7616
HasCollege	-0.2794	0.4437	-0.63	0.529	0.7562
IsProfessional	0.2253	0.465	0.49	0.628	1.2527
IsRetired	-1.159	0.9323	-1.24	0.214	0.3138
Unemployed	0.9886	4.69	0.21	0.833	2.6875
ResidenceLength	0.02468	0.0138	1.79	0.0738	1.025
DualIncome	0.4518	0.5215	0.87	0.3863	1.5711
Minors	1.133	0.4635	2.44	0.0145	3.105
Own	1.056	0.5594	1.89	0.059	2.8748
House	-0.9265	0.6218	-1.49	0.1362	0.3959
White	1.864	0.5454	3.42	0.0006	6.4495
English	1.53	0.8407	1.82	0.0687	4.6182
PrevChildMag	1.557	0.7119	2.19	0.0287	4.7446
PrevParentMag	0.4777	0.624	0.77	0.4439	1.6124

Logistic Regression - errors

- **Deviance -** the discrepancy between **the current model** and **the full model**. The full model is the model that has n parameters, one parameter per observation.
- **Likelihood Ratio** **LR = *-2ln(L(m1) / L(m2))***, where m1,m2 – two models, L – Likelihood.
- **Wald Test** - approximates the LR test, but tests only one model against the null hypothesis that a set of parameters is equal to some value.
- **Lagrange Multiplier test (Score)** - estimating only a single model, like Wald test. The test statistics is the expected change in the chi-squared statistic for the model if a variable or set of variables is added to the model. The Lagrange multiplier test is sometimes also referred to as a test for omitted variables. They are also sometimes referred to as modification indices.



The likelihood is the probability of the data given the parameter estimates (model).

The goal is to find values for the parameters (coefficients) of the model that maximize value of the likelihood function, that is, to find the set of parameter estimates that make the data most likely.

It is easier to work with the log of the likelihood, rather than the likelihood itself, because it is easier to work with.
The log likelihood (i.e., the log of the likelihood) will always be negative, with higher values (closer to zero) indicating a better fitting model.

Wald test
- https://en.wikipedia.org/wiki/Wald_test

Logistic Regression – Standard Errors

Fitting the model is done using **MLE** (Maximum Likelihood Estimation) method.

Fitting means - find the vector of parameters $\vec{\beta}$ that maximizes the conditional probability of observing the training data.

The MLE solution can be found by **gradient descent**.

The **standard errors** of model coefficients are the **square roots** of the **diagonal entries** of the **covariance matrix** of $\vec{\beta}$.
- https://www.efavdb.com/logistic-regression

Covariance matrix (also known as auto-covariance matrix, dispersion matrix, variance matrix, or variance–covariance matrix) is a square matrix giving the covariance between each pair of elements of a given random vector.

Diagonal elements of this matrix are variances, i.e., the covariance of each element with itself.

Generally **it is not possible to write down an explicit expression** for the elements in this matrix (except for very special cases).

MLE theory states that the estimated large-sample variance-covariance matrix for MLE is given by the **inverse of the information matrix**.
The elements of **Information Matrix** are the negatives of the expected values of the 2nd order partial derivatives of the logarithm of the likelihood function evaluated at $\vec{\beta} = \vec{b}$.

See, for example, this book: Linear Statistical Inference and its Applications - by C. Radhakrishna Rao , 1973

covariance matrix is given by the **inverse of the information matrix**.
The elements of **Information Matrix** are the negatives of the expected values of the 2nd order partial derivatives of the logarithm of the likelihood function evaluated at $\vec{\beta} = \vec{b}$.

$$\left[-E\left\{\frac{\partial^2 \log_e L(\boldsymbol{\beta})}{\partial\beta_i\,\partial\beta_j}\right\}_{\boldsymbol{\beta}=\mathbf{b}}\right]^{-1}$$

$$\mathrm{cov}(\vec{\beta}^*, \vec{\beta}^*) \sim H(J)^{-1}$$

$$\approx \frac{1}{\sum_k \vec{x}_k \vec{x}_k^T p(y=0|\vec{x}_k) p(y=1|\vec{x}_k)}$$

$$\sigma^2_{\beta_i^*} = \mathrm{cov}(\vec{\beta}^*, \vec{\beta}^*)_{ii}.$$

$$p(y=1|\vec{x}) \in \left(\frac{1}{1+e^{-\vec{x}\cdot\vec{\beta}^* + z\sigma_{\vec{x}\cdot\vec{\beta}^*}}}, \frac{1}{1+e^{-\vec{x}\cdot\vec{\beta}^* - z\sigma_{\vec{x}\cdot\vec{\beta}^*}}}\right)$$

Logistic Regression – Covariance Matrix calculation (Python)

https://stats.stackexchange.com/questions/89484/how-to-compute-the-standard-errors-of-a-logistic-regressions-coefficients

```
import numpy as np
from sklearn import linear_model

logit = linear_model.LogisticRegression()
resLogit = logit.fit(X_train, y_train)
predProbs = resLogit.predict_proba(X_train)

# Design matrix -- add column of 1's at the beginning of your X_train matrix
X_design = np.hstack([np.ones((X_train.shape[0], 1)), X_train])

# Initiate matrix of 0's, fill diagonal with each predicted observation's variance
V = np.diagflat(np.product(predProbs, axis=1))

# Covariance matrix (uses @-operater for matrix multiplication in Python 3.5+)
covLogit = np.linalg.inv(X_design.T @ V @ X_design)
print("Covariance matrix: ", covLogit)

# Standard errors
print("Standard errors: ", np.sqrt(np.diag(covLogit)))

# Wald statistic (coefficient / s.e.) ^ 2
logitParams = np.insert(resLogit.coef_, 0, resLogit.intercept_)
print("Wald statistics: ", (logitParams / np.sqrt(np.diag(covLogit))) ** 2)
```

$(X^T V X)^{-1}$

Central Limit Theorem (CLT)

We have all heard about **Normal (Gaussian) distribution**.

This distribution is very foundational in statistics. And **it comes from CLT**.

Classical CLT mathematically proves that if we calculate group averages in many groups, then the distribution of those averages converges to Normal distribution as long as we have large number of points in each group (30..100 or more) and large number of groups (30-100 or more).

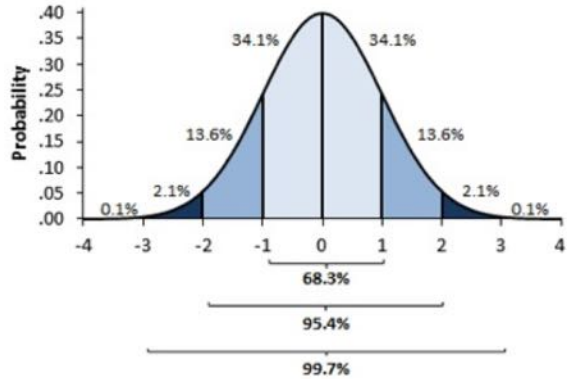
Specifically, suppose we have our data points sampled from a distribution with finite average and finite variance. We subdivide this data into Ng groups, each containing Ns samples:

[...] [...] [...] [...] [...] [...] [...] [...] [...]

For each group we calculate average value of its Ns samples. Thus we have Ng averages values, and we plot them as a histogram.

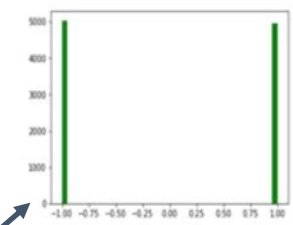
If Ng and Ns are big enough, the histogram will converge to a normal distribution. This holds true regardless of the shape of the original distribution of data.

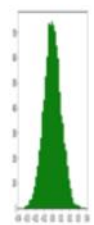
For example, suppose we have a million data points, each value is either 1 or -1 (random coin toss). We split data into Ng=10,000 groups, each sample group has Ns=10,000 data points. Then in each group we calculate the mean value (which will be close to zero).
Now we have 10,000 of mean values - and we make a histogram of those values. According to CLT it will be bell-shaped.



$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$\mu =$ Mean
$\sigma =$ Standard Deviation
$\pi \approx 3.14159\cdots$
$e \approx 2.71828\cdots$





Expected value of abs(x) : $\sigma\sqrt{\frac{2}{\pi}}$

Expected value of x^2 : $\boldsymbol{\sigma^2}$

$$\mathrm{E}[(X-\mu)^p] = \begin{cases} 0 & \text{if } p \text{ is odd,} \\ \sigma^p (p-1)!! & \text{if } p \text{ is even.} \end{cases}$$

Here double factorial "n!!" denotes the product of all numbers from n down to 1 that have the same parity as n.
E.g. 9!! = 9*7*5*3*1, 6!! = 6*4*2

History of Normal Distribution and CLT

The formula for the normal distribution was discovered independently by **Adrain** in 1808 and **Gauss** in 1809.

In **1809 Gauss** published his monograph "Theoria motus corporum coelestium in sectionibus conicis solem ambientium" where among other things he introduces several important statistical concepts, such as the **method of least squares**, the **method of maximum likelihood**, and the **normal distribution**.

Earlier **Laplace** in **1774** posed the problem of aggregating several observations. His own solution led to the Laplacian distribution (exponential).

It was **Laplace** who first calculated the value of the integral $\int e^{-t^2}\,dt = \sqrt{\pi}$ in **1782**, providing the **normalization constant** for the normal distribution.

Finally, it was **Laplace** who in **1810** proved and presented to the Academy the fundamental **Central Limit Theorem**, which emphasized the theoretical importance of the normal distribution.

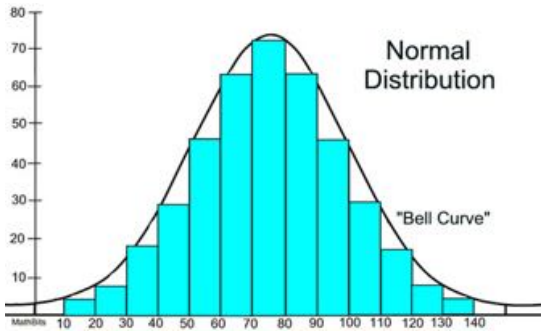
Gauss and Laplace are very famous. The normal distribution was called for many years **"Gaussian-Laplace distribution"**.

Robert Adrain is less known. He was a brilliant mathematician. His main contributions are
- the **method of least squares**
- and derivation of the **formula for normal distribution**.

Pierre-Simon Laplace
1749–1827
France

Robert Adrain
1775–1843
USA

Carl Friedrich Gauss
1777 – 1855
from Göttingen,
Germany



p-value & null hypothesis

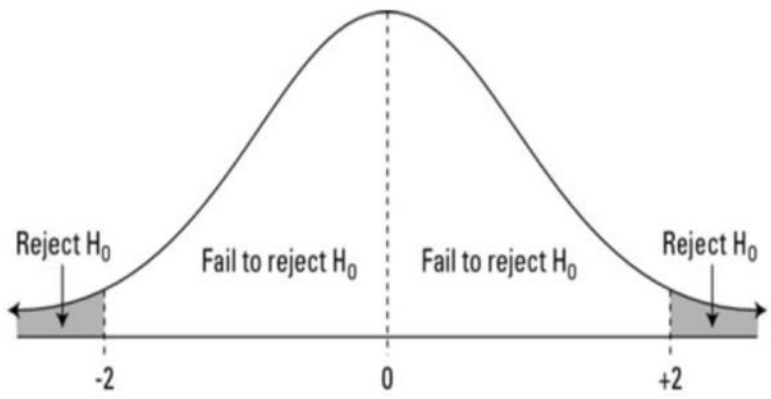
A **p-value** (probability value) is a probability of getting results at least as extreme as yours. If this probability is very small, this means that the result is unlikely to be explained by the null hypothesis. So we should reject the null hypothesis.

```
P-value    probability of observed
-------------------------------
P < 0.05         5%
P < 0.01         1%
p < 0.001        0.1%
-------------------------------
```

A small p-value (typically ≤ 0.05) indicates strong evidence against the null hypothesis, so you reject the null hypothesis. People say that this evidence is "statistically significant".

Related term is **Confidence Interval (CI)**. A 95% CI is the interval that you are 95% certain contains the true population value as it might be estimated from a much larger study.

z-score is the distance from the mean measured in the number of standard deviations.



Type I error - false rejection of the null hypothesis
Type II error - false acceptance of the null hypothesis

\ Decision Truth \	Decision to Accept Ho	Decision to Reject Ho
Ho is True	Correct Decision	type I error
Ho is False	type II error	Correct Decision

Problem0:

We tossed a coin 10 times and got 8 heads. Compute p-value.
To compute the p-value we just need compute the probability of 8 or more heads in 10 tosses (assuming the coin is fair).

Note that the number of heads follows a binomial distribution with n=10 and p=0.5.

So the p-value = P[8] + P[9] + P[10] = .044 + .01 + .001 = 0.055.

Problem1:

Given: women's heights are normally distributed with the mean 65" and standard deviation of 3.5"
Question: What is the probability of finding a random sample of 50 women with a mean height of 70".

To answer, let's estimate the **z-score**:
$z = (x - \mu) / (\sigma / \sqrt{n}) = (70 - 65) / (3.5/\sqrt{50}) = 5 / 0.495 = 10.1$

We know that 99% of values fall within 3σ (68, 95, 99.7 rule).
We have z-score = 10, which is much more than 3.
So the probability that any sample of 50 women will have a mean height of 70" will be much-much less than 1%.

Problem 2:

Suppose you are testing a claim that the percentage of all women with varicose veins is 25%, and your sample of 100 women had 20% with varicose veins.
Then the sample proportion p=0.20.
The standard error for your sample percentage is the square root of p(1-p)/n which equals 0.04 or 4%.

You find the test statistic by taking the proportion in the sample with varicose veins, 0.20, subtracting the claimed proportion of all women with varicose veins, 0.25, and then dividing the result by the standard error, 0.04.

These calculations give you a test statistic (standard score) of –0.05 divided by 0.04 = –1.25.

This tells you that your sample results and the population claim in H0 are 1.25 standard errors apart; in particular, your sample results are 1.25 standard errors below the claim.

When testing H0: p = 0.25 versus Ha: p < 0.25, you find that the p-value of -1.25 by finding the probability that Z is less than -1.25. When you look this number up on the **Z-table**, you find a probability of 0.1056 of Z being less than this value.

Note: A **Z-table** is simply a table of values of the cumulative distribution function of the normal distribution.

The dichotomous use of p-value in statistical inference is worse than useless.

1. p-value is a conditional probability, i.e., **Pr(Data | H0 is true)**
   but what we want to know is **Pr(H0 is true | Data)**
2. the calculation of p-value depends on at least three things:
   - the effect size (e.g., difference of two means due to two treatment, odds ratio, etc.),
   - the uncertainty level of the point estimate (e.g., standard deviation),
   - and sample size
3. it is a standard result in statistics theory that, based on one set of sample data we cannot tell the observed effect **size is due to treatment or due to random variation** (i.e., sampling distribution requires we repeat the same experiment many times before any confirmatory conclusion can be made)
4. if by nature the H0 is false (in the majority cases of real life data / research questions), p-value =0.

In summary, P-value based Null Hypothesis Significance Test (NHST) is logically not defensible, technically flawed and practically damaging

Gang (John) Xie,
Ph.D. in Statistics,
Quantitative Consulting Unit in Research Office

https://www.researchgate.net/post/What_is_the_relationship_between_R-squared_and_p-value_in_a_regression

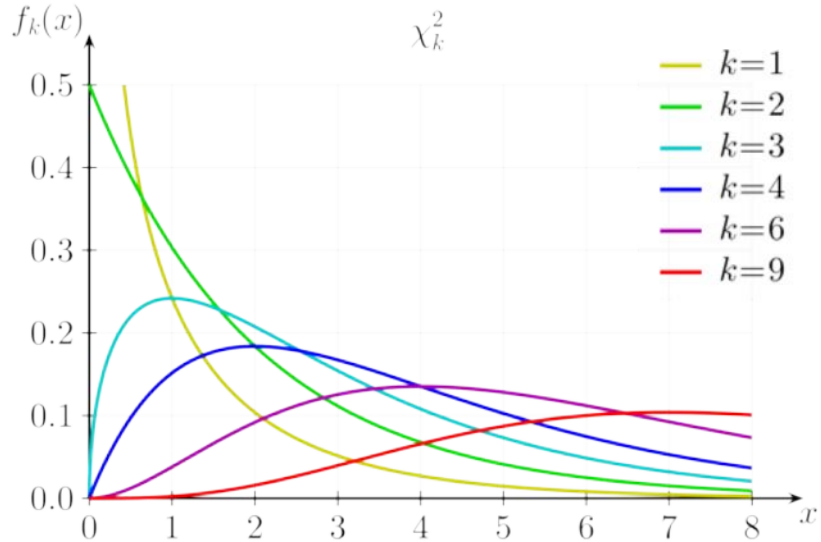
Chi-Square χ^2 Distribution

If $Z_1, \ldots, Z_k$ are independent, standard normal (Gaussian) random variables, then the sum of their squares:

$$\text{sum}_k(Z_i^2)$$

is distributed as χ^2, with "k" degrees of freedom

Mean = K
variance = 2K

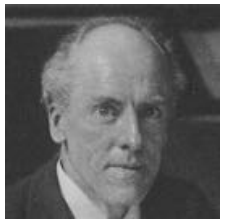


$$f(x;\, k) = \begin{cases} \dfrac{x^{\frac{k}{2}-1} e^{-\frac{x}{2}}}{2^{\frac{k}{2}} \Gamma\left(\frac{k}{2}\right)}, & x > 0; \\ 0, & \text{otherwise.} \end{cases}$$

Why is it called $\chi 2$?

Greek character Chi (χ) appears to be the notation used by Karl Pearson in 1900:

"On the Criterion that a Given System of Deviations from the Probable in the Case of a Correlated System of Variables is such that it can be Reasonably Supposed to have Arisen from Random Sampling," Philosophical Magazine, 50, (1900), 157-175.



Karl Pearson
1857 – 1936
English mathematician
and biostatistician

Where Gamma Function:

$$\Gamma(n) = (n-1)!$$

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x}\, dx, \qquad \Re(z) > 0$$

Chi-Square χ^2 Goodness of Fit Test

https://stattrek.com/chi-square-test/goodness-of-fit.aspx

Pearson, 1900.
Suppose we have k mutually-exclusive classes
N - total number of all observations,
n_i (i=1,2,...k) - number of observations for each class

p_i - hypothetical probability of each class (null hypothesis).
So sum(p_i) = 1

Then for each class:
expected value of observations m_i=N*p_i
sum(m_i) = N*sum(p_i) = N = sum(n_i)

Karl Pearson has proposed that as N gets bigger, the value

X^2 = sum((n_i-m_i)^2/m_i) --> **χ^2 distribution with (k – 1) degrees of freedom**

Notation

χ2 is the lower-case Greek letter Chi
f_o is the Observed Frequency
f_e is the Expected Frequency

Chi-Square Statistic

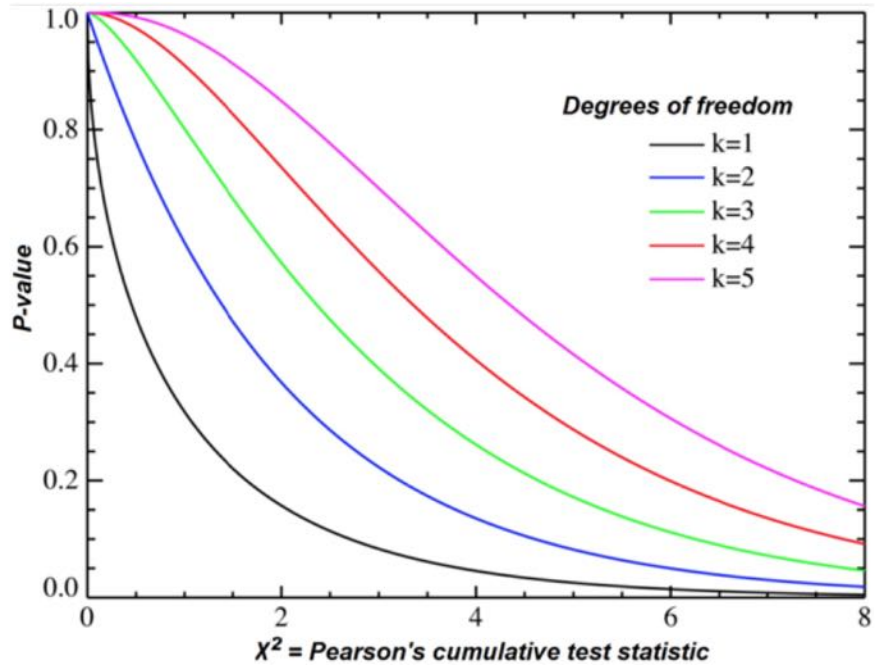
$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}$$

Computing Expected Frequencies

fc - frequency total for the column
fr - frequency total for the row

$$f_e = \frac{f_c f_r}{n}$$

Chi-squared test, showing χ2 on the x-axis and p-value on the y-axis.



In practice we calculate value X^2 from our data, then use χ^2 distribution to estimate the P-value (the probability that a chi-square statistic is more extreme than X^2 value that we have.

Example of χ^2 test (from https://stattrek.com/chi-square-test/goodness-of-fit.aspx)

Problem:

We have 3 types of cards (T1,T2,T3) with theoretical frequency of (30%, 60%, 10%).
We have a random sample of 100 cards with counts (50, 45, 5).
Is this sample consistent with theory? Use a 0.05 level of significance.

Solution:

Null hypothesis: The proportions are (30%, 60%, 10%)
Alternative hypothesis: At least one of the proportions in the null hypothesis is false.
Degrees of Freedom DF = k-1 = 3-1 = 2
Expected values:
$E_1 = 100*0.3 = 30$, $E_2 = 100*0.6 = 60$, $E_3 = 100 * 0.10 = 10$
$X^2 = \Sigma [(O_i - E_i)^2 / E_i] = [(50 - 30)^2 / 30] + [(45 - 60)^2 / 60] + [(5 - 10)^2 / 10] = 19.58$

The P-value is the probability that a chi-square statistic having 2 degrees of freedom is more extreme than 19.58.
Using Chi-Square Distribution we find P(X2 > 19.58) = 0.0001.

Since the P-value (0.0001) is less than the significance level (0.05), we reject the null hypothesis.

Note: If you use this approach on an exam, you may also want to mention why this approach is appropriate.
Specifically, the approach is appropriate because the sampling method was simple random sampling, the variable under study was categorical, and each level of the categorical variable had an expected frequency count of at least 5.

Proof and Intuition for Chi-squared test

Seven proofs of the Pearson Chi-squared independence test and its graphical interpretation – by Eric Benhamou & Valentin Melot , 2018.
- https://arxiv.org/pdf/1808.09171.pdf

Example: we have black and white balls.
The null hypothesis is that the probability of white balls is p (and black – (1-p)).
You have drawn a sample of n balls and observed k white balls.
The exact binomial probability of obtaining k white balls from n is

$P(K=k \mid n,p)=\mathrm{bin}(n,k)p^k(1-p)^{n-k}$, where bin(n,k) is the binomial coefficient.

Now you can approximate the probability by calculating the z-score.
The mean of the binomial distribution is nq
and standard deviation s = sqrt(np(1−p)).
So the z-score is z=(k−np)/s and it follows standard normal distribution.

On the the hand, a squared normally distributed variable (or their sum) follows a $\chi 2$-distribution.
So, in this case the square z^2 is $\chi 2$-distributed (with 1 degree of freedom).

Note that z^2 can be expressed as $\frac{(k-nq)^2}{nq} + \frac{(n-k-n(1-q))^2}{n(1-q)}$

which is a more familiar expression for the $\chi 2$-statistic.

for white balls:
np = expected value
k = observed value
for black balls:
n(1-p) = expected value
n-k = observed value

Asymptotical proof

if k=2

$$T = \frac{(X_1 - np_1)^2}{np_1} + \frac{(X_2 - np_2)^2}{np_2}$$
$$= \frac{(X_1 - np_1)^2}{np_1} + \frac{(n - X_1 - n[1 - p_1])^2}{n(1 - p_1)}$$
$$= \frac{(X_1 - np_1)^2}{np_1} + \frac{(X_1 - np_1)^2}{n(1 - p_1)}$$
$$= \frac{(X_1 - np_1)^2}{np_1(1 - p_1)} \quad \text{since} \quad \frac{1}{p} + \frac{1}{1-p} = \frac{1}{p(1-p)}$$

X_1 is a binomial random variable with mean np_1 and variance $np_1(1 - p_1)$

$$\frac{(X_1 - np_1)}{\sqrt{np_1(1 - p_1)}} \sim N(0,1)$$

$$\left(\frac{(X_1 - np_1)}{\sqrt{np_1(1 - p_1)}}\right)^2 \sim X_{,1}^2$$

Chi-Square χ^2 Test for Independence

(two-way tables, association)

Evaluates the relationship between two variables.
It is a non-parametric test, performed on categorical data.

Example:
500 boys and girls are asked which is their favorite color: blue, green, or pink (see results).
Using alpha = 0.05, would you conclude that there is a relationship between gender and favorite color?

Degrees of Freedom df = (rows-1)(columns-1) = (2-1)(3-1) = 2

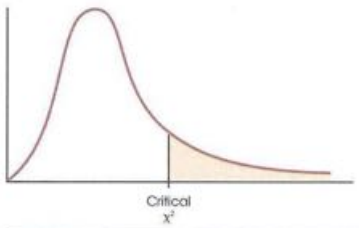
In χ^2 table the value for alpha=0.05 and df=2 is 5.99

Calculate statistics: χ^2 = sum $(f_o - f_e)^2 / f_e$ = 276.39 >> 5.99

Conclusion: yes, there is a relationship

OBSERVED	Blue	Green	Pink	
Boys	100	150	20	300
Girls	20	30	180	200
	120	180	200	N = 500

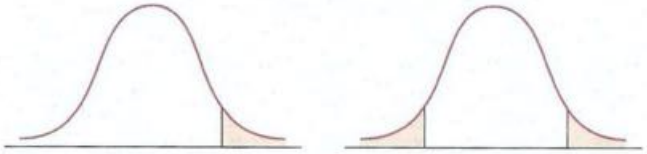
EXPECTED	Blue	Green	Pink	
Boys	72	108	120	300
Girls	48	72	80	200
	120	180	200	N = 500



χ^2

df	Proportion in Critical Region 0.10	0.05	0.025	0.01	0.005
1	2.71	3.84	5.02	6.63	7.88
2	4.61	5.99	7.38	9.21	10.60
3	6.25	7.81	9.35	11.34	12.84
4	7.78	9.49	11.14	13.28	14.86
5	9.24	11.07	12.83	15.09	16.75
6	10.64	12.59	14.45	16.81	18.55
7	12.02	14.07	16.01	18.48	20.28
8	13.36	15.51	17.53	20.09	21.96
9	14.68	16.92	19.02	21.67	23.59
10	15.99	18.31	20.48	23.21	25.19
11	17.28	19.68	21.92	24.72	26.76
12	18.55	21.03	23.34	26.22	28.30
13	19.81	22.36	24.74	27.69	29.82
14	21.06	23.68	26.12	29.14	31.32
15	22.31	25.00	27.49	30.58	32.80
16	23.54	26.30	28.85	32.00	34.27
17	24.77	27.59	30.19	33.41	35.72
18	25.99	28.87	31.53	34.81	37.16
19	27.20	30.14	32.85	36.19	38.58
20	28.41	31.41	34.17	37.57	40.00
21	29.62	32.67	35.48	38.93	41.40
22	30.81	33.92	36.78	40.29	42.80
23	32.01	35.17	38.08	41.64	44.18
24	33.20	36.42	39.36	42.98	45.56
25	34.38	37.65	40.65	44.31	46.93
26	35.56	38.89	41.92	45.64	48.29
27	36.74	40.11	43.19	46.96	49.64
28	37.92	41.34	44.46	48.28	50.99
29	39.09	42.56	45.72	49.59	52.34
30	40.26	43.77	46.98	50.89	53.67
40	51.81	55.76	59.34	63.69	66.77
50	63.17	67.50	71.42	76.15	79.49
60	74.40	79.08	83.30	88.38	91.95
70	85.53	90.53	95.02	100.42	104.22
80	96.58	101.88	106.63	112.33	116.32
90	107.56	113.14	118.14	124.12	128.30
100	118.50	124.34	129.56	135.81	140.17

t



df	Proportion in One Tail 0.25	0.10	0.05	0.025	0.01	0.005
	Proportion in Two Tails Combined 0.50	0.20	0.10	0.05	0.02	0.01
1	1.000	3.078	6.314	12.706	31.821	63.657
2	0.816	1.886	2.920	4.303	6.965	9.925
3	0.765	1.638	2.353	3.182	4.541	5.841
4	0.741	1.533	2.132	2.776	3.747	4.604
5	0.727	1.476	2.015	2.571	3.365	4.032
6	0.718	1.440	1.943	2.447	3.143	3.707
7	0.711	1.415	1.895	2.365	2.998	3.499
8	0.706	1.397	1.860	2.306	2.896	3.355
9	0.703	1.383	1.833	2.262	2.821	3.250
10	0.700	1.372	1.812	2.228	2.764	3.169
11	0.697	1.363	1.796	2.201	2.718	3.106
12	0.695	1.356	1.782	2.179	2.681	3.055
13	0.694	1.350	1.771	2.160	2.650	3.012
14	0.692	1.345	1.761	2.145	2.624	2.977
15	0.691	1.341	1.753	2.131	2.602	2.947
16	0.690	1.337	1.746	2.120	2.583	2.921
17	0.689	1.333	1.740	2.110	2.567	2.898
18	0.688	1.330	1.734	2.101	2.552	2.878
19	0.688	1.328	1.729	2.093	2.539	2.861
20	0.687	1.325	1.725	2.086	2.528	2.845
21	0.686	1.323	1.721	2.080	2.518	2.831
22	0.686	1.321	1.717	2.074	2.508	2.819
23	0.685	1.319	1.714	2.069	2.500	2.807
24	0.685	1.318	1.711	2.064	2.492	2.797
25	0.684	1.316	1.708	2.060	2.485	2.787
26	0.684	1.315	1.706	2.056	2.479	2.779
27	0.684	1.314	1.703	2.052	2.473	2.771
28	0.683	1.313	1.701	2.048	2.467	2.763
29	0.683	1.311	1.699	2.045	2.462	2.756
30	0.683	1.310	1.697	2.042	2.457	2.750
40	0.681	1.303	1.684	2.021	2.423	2.704
60	0.679	1.296	1.671	2.000	2.390	2.660
120	0.677	1.289	1.658	1.980	2.358	2.617
∞	0.674	1.282	1.645	1.960	2.326	2.576

Student's *t*-distribution

https://en.wikipedia.org/wiki/Student%27s_t-distribution

Student's t-distribution (or simply the t-distribution)
is any member of a family of continuous probability distributions
that arises when **estimating the mean of a normally distributed population**
in situations **where the sample size is small**
and the population standard deviation is unknown.
It was developed by **William Sealy Gosset** (1908)
under the pseudonym **Student**.

Let Xi be independently and identically drawn from the normal distribution with expected mean value **μ** and variance **σ²**

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

----- Sample mean

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

----- Sample variance

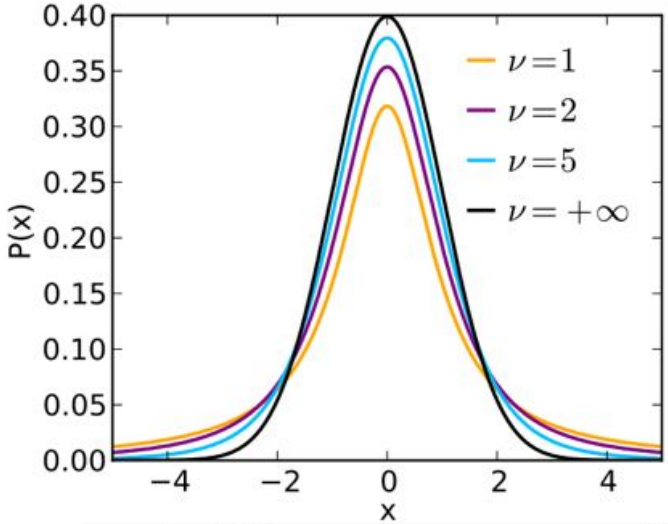
$$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$$

----- Random variable, Normal distribution

$$\frac{\bar{X}-\mu}{S/\sqrt{n}};$$

----- Random variable, **t-distribution**. It uses sample variance (because real sigma is unknown). This creates **uncertainty** – and leads to the "**fat tails**" in the t-distribution.

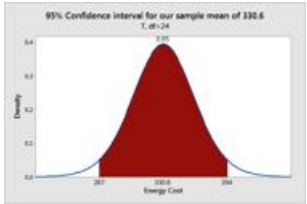
William Sealy Gosset used pseudonym "Student" because he was working for **Guinness Beer Company** in England as a "**Head Experimental Brewer**" (working on cultivating barley), and the company didn't allow him to publish under his own name.



$$\frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)}\left(1+\frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

In Excel: T.DIST(x,DF,FALSE)
where DF = Degrees of Freedom (Npoints-1)

95% confidence interval



t-statistics vs z-statistics

z-statistics:

When sample size is big (> 30), then in many cases sample distribution is close to gaussian. So we use gaussian distribution and z-statistics.

Suppose that variable x follows normal gaussian distribution with mean value **μ** and standard deviation **σ**. Then we can compose variable "z" which would follow standard normalized distribution with zero mean and standard deviation = 1 like this:

$$z = (x-\mu)/\sigma$$

Z-score is simply the z-value which shows how far is the value from the mean expressed in number of "sigmas". For example, if z > 2, that means that the original value is more than two sigmas away from the mean value.

For normal distribution this means that the probability of this event is less than 0.05.
(to be precise, the exact z-score for 0.05 is 1.96)

t-statistics:

When sample size is small (N < 30), then even if we know that the underlying distribution is gaussian, we still can not evaluate the distribution's sigma with enough precision. In this case we need to use t-distribution and t-statistics.

$$t = (x-\mu)/s$$

Here **s** is a sample mean.

t-distribution has "fat tails", so to get 95% confidence interval we need to use t-score larger than 2. How much larger depends on how many data points we have in the sample. **df = degrees of freedom = N-1**

```
N-1    95% t-score
------------------
 1   : 12.71
 2   :  4.30
 3   :  3.18
 5   :  2.57
10   :  2.23
20   :  2.09
30   :  2.04
60   :  2.00
------------------
```

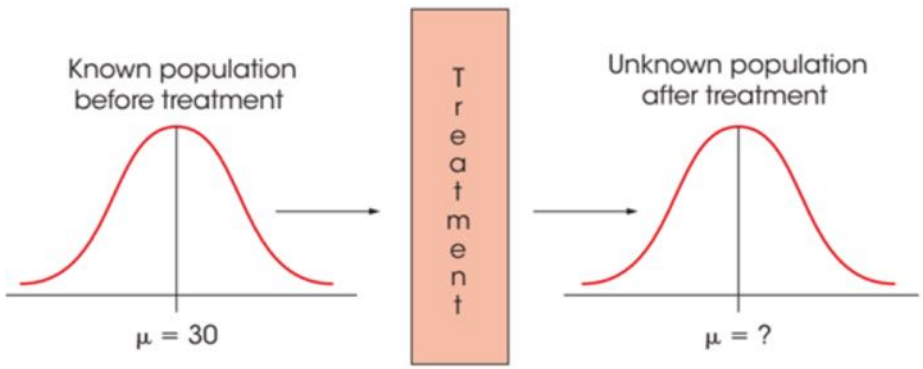
```
from scipy import stats
# Student, p<0.05, 2-tail, same as Excel TINV(0.05,n-1)
for df in [1,2,3,5,10,20,30,60]:
    t_score = stats.t.ppf(1-0.025, df)
    print(f"{df:2}   : {t_score:5.2f}")
```

Hypothesis Tests with the *t* Statistic

The one-sample t test statistic (assuming the null hypothesis is true)

$$t = \frac{\text{sample mean} - \text{population mean}}{\text{estimated standard error}} = \frac{M - \mu}{s_M} = 0$$

Basic Research Situation for t Statistic:



Assumed that μ is known for the population before treatment.
The purpose of the research study is to determine whether the treatment has an effect.
Note that the population after treatment has unknown values for the mean and the variance. So we will use a sample to test a hypothesis about the population mean.

Four (4) Steps of Hypothesis Testing:

1. State the null hypotheses (H0) and alternative hypotheses, and select an alpha level
2. Locate the critical region using the t-distribution table and df
3. Calculate the t test statistic
4. Make a decision regarding H0 (null hypothesis)

For example:

1. H0: no effect (same mean), two-tailed test, alpha = 0.01
2. For sample size 100, critical t-values = +/-2.6264
3. t-statistics = (sample_mean – population_mean) / s , where s is an estimated standard error (calculated from sample)
4. if || t-statistics || > critical value, then we reject the H0

t-test – continued

Effect size.

In addition to rejecting/accepting a Hypothesis, you can also estimate the size of the effect:

- **Cohen's d** - for comparing two samples/groups with same sizes and similar SDs.
- **Glass's delta** - uses only the SD of the control group.
- **Hedges' g** - for two groups with different sample sizes.

Example: for two groups with mean M and standard deviations SD :

Cohen's d = $(M_2 - M_1)/SD_{pooled}$

where

$SD_{pooled} = \sqrt{((SD_1^2 + SD_2^2)/2)}$

Determining the amount of variability in scores explained by the treatment effect is an alternative method for measuring effect size.

$$r^2 = \frac{\text{variability accounted for}}{\text{total variability}} = \frac{t^2}{t^2 + df}$$

r2 = 0.01 small effect
r2 = 0.09 medium effect
r2 = 0.25 large effect

Estimating m from sample

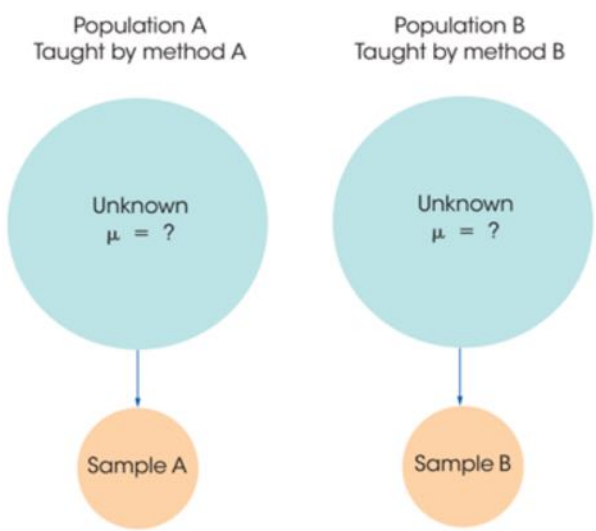
(use sample mean and sample deviation)

$$t = \frac{M - \mu}{s_M} \quad \Rightarrow \quad \mu = M \pm t s_M$$

For any α we know that (1 – α) proportion of t values fall between ± t for the appropriate df.

The t Test for Two Independent Samples

Data comes from two completely different, independent groups.
Observations within each sample must be independent.
The two populations must be normal.
The two populations must have equal variances (called homogeneity of variance).



Null hypothesis for independent-measures test:

$$H_0 : \mu_1 - \mu_2 = 0$$

Basic structure of the t statistic

$$t = \frac{(M_1 - M_2) - (\mu_1 - \mu_2)}{s_{(M_1 - M_2)}}$$

where estimated standard error (unbiased if n1=n2)

$$s_{(M_1 - M_2)} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

unbiased estimate:

$$s_p^2 = \frac{SS_1 + SS_2}{df_1 + df_2}$$

SS = Sum of Squares
Degrees of freedom df = $df_1 + df_2 = (n_1 - 1) + (n_2 - 1)$

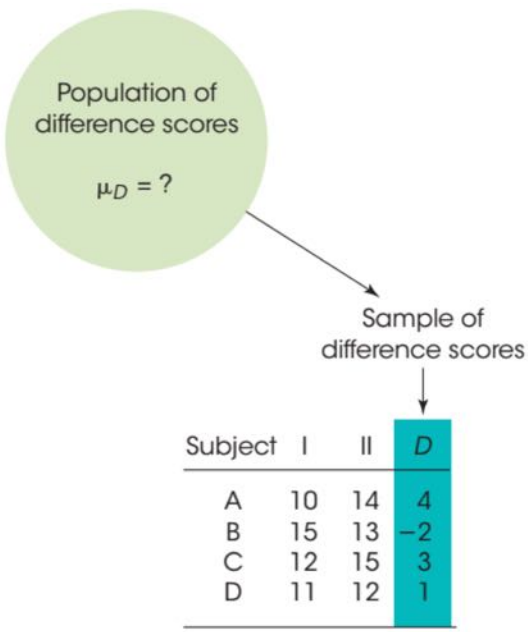
Test for homogeneity of variance:
Levene's test or Brown–Forsythe test
https://en.wikipedia.org/wiki/Levene%27s_test

The t Test for Repeated-Measures

Repeated-measures design
(a.k.a. within-subjects design)

Two separate scores are obtained **for each individual in the sample**

Same subjects are used in both groups.
Observations within each treatment condition must be independent.
Population distribution of difference scores (D values) must be normally distributed.



Subject	I	II	D
A	10	14	4
B	15	13	−2
C	12	15	3
D	11	12	1

Difference score = $D = X_2 - X_1$

Mean Difference: $M_D = \frac{\sum D}{n}$

Null Hypothesis H_0: $\mu_D = 0$

$$t = \frac{M_D - \mu_D}{s_{M_D}}$$

$$\text{variance of D} = s^2 = \frac{SS}{df}$$

$$s_{M_D} = \sqrt{\frac{s^2}{n}}$$

$$estimated\ Cohen's\ d = \frac{M_D}{s}$$

$$r^2 = \frac{t^2}{t^2 + df}$$

Confidence Interval (C.I.):

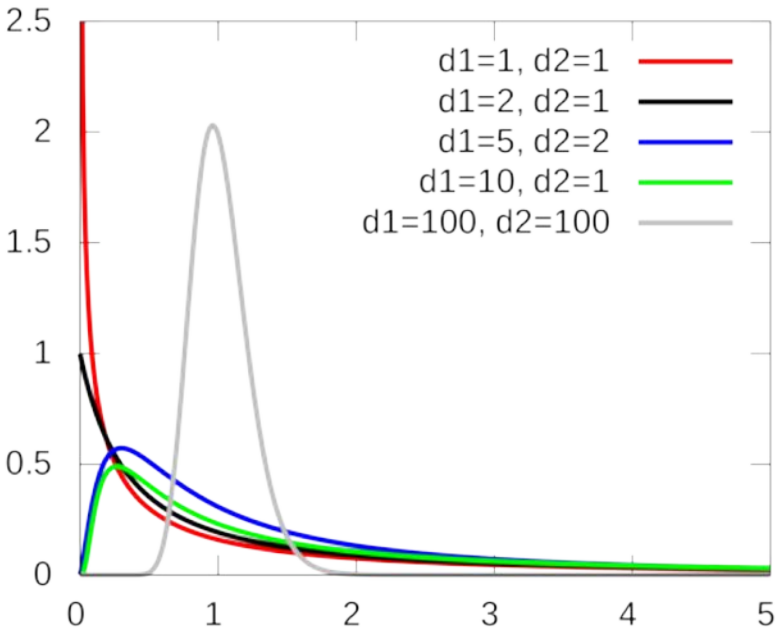
$$C.I.(1-\alpha): \mu_D = M_D \pm t_\alpha s_{M_D}$$

F-Distribution (Fisher–Snedecor)

F-distribution
- right-skewed distribution used most commonly in Analysis of Variance.
- (https://en.wikipedia.org/wiki/F-distribution)
- named after Ronald Fisher and George W. Snedecor

Ronald Fisher 1890-1962
British. Developed the
analysis of variance ANOVA

George W. Snedecor
1881-1974, USA



A random variate of the F-distribution with parameters d1, d2 arises as the ratio of two appropriately scaled chi-squared variates:

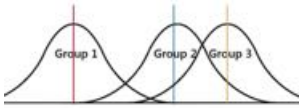
$$X = (U_1/d_1) / (U_2/d_2)$$

where U_1 and U_2 are independent and have chi-squared distributions with d_1 & d_2 degrees of freedom respectively.

F-tests:

where the null hypothesis is that two independent normal variances are equal,
and the observed sums of some appropriately selected squares are then examined to see whether their ratio is significantly incompatible with this null hypothesis.

ANOVA = Analysis of Variance



Ronald Fisher 1890-1962
British. Developed the analysis of variance **ANOVA**

ANOVA - a collection of statistical models and estimation procedures (such as the "variation" among and between groups)
used to analyze the differences among group means in a sample.

The observed variance in a particular variable is partitioned into components attributable to different sources of variation.

In its simplest form, **ANOVA** provides a statistical test of whether two or more population means are equal, and therefore generalizes the t-test beyond two means.

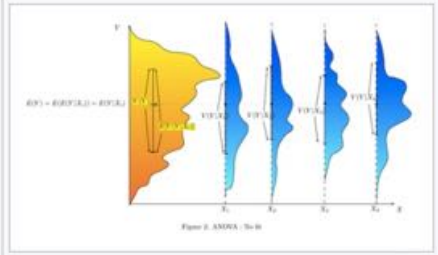
Variance Between

Variance Within

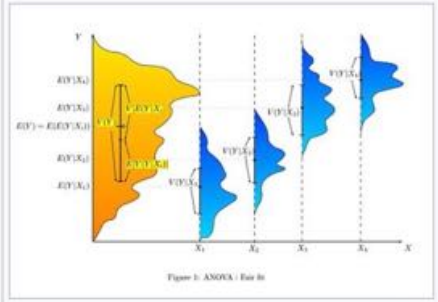
ANOVA is the synthesis of several ideas and it is used for multiple purposes.
As a consequence, it is difficult to define concisely or precisely.

"Classical" **ANOVA** for balanced data does three things at once:
1. Uses additive data decomposition for exploratory data analysis. Uses sums of squares to indicate the variance of each component of the decomposition.
2. Comparisons of mean squares, along with an F-test ... allow testing of a nested sequence of models.
3. Closely related to the **ANOVA** is a linear model fit with coefficient estimates and standard errors.

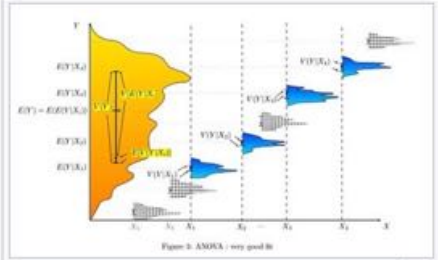
Additionally, **ANOVA** :
1. is computationally elegant and relatively robust against violations of its assumptions.
2. provides strong (multiple sample comparison) statistical analysis.
3. has been adapted to the analysis of a variety of experimental designs.



No fit: Young vs old, and short-haired vs long-haired



Fair fit: Pet vs Working breed and less athletic vs more athletic



Very good fit: Weight by breed

Fisher Exact Test

From Fisher's book "The Design of Experiments" (1935)

Lady Tasting Tea example in 2nd chapter

Lady claimed that she ca distinguish between two types of cups of tea:

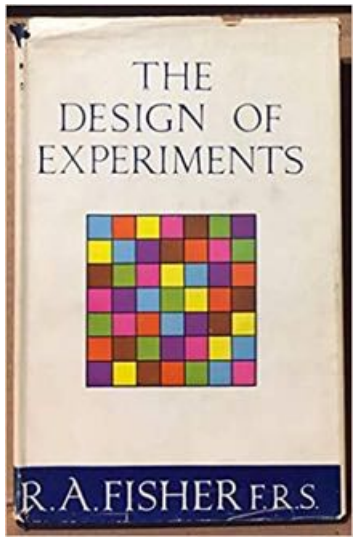
1. tea added first, then milk
2. milk added first, then tea

Author introduces the concept of "null hypothesis"

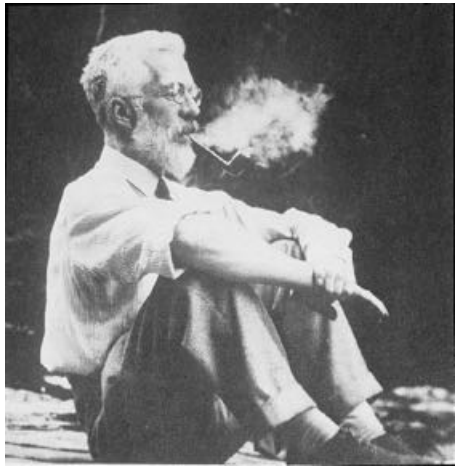
Experiment: given 4 cups of type 1 and 4 cups of type 2.
They look the same.
Lady should test them and label the "milk first" ones.

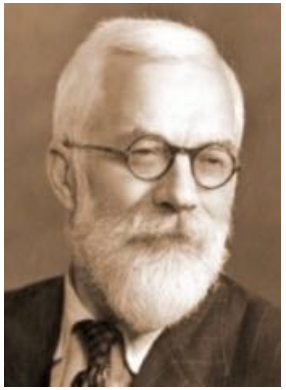
The test is called "exact" because Fisher demonstrates how in this case he can calculate exact probability of each outcome.

If she guessed 3 out of 4, the probability is 0.22 – not good
If she guessed 4 out of 4 – the probability is 1/70 < 0.05 - good



1935





Hoeffding's inequality

https://en.wikipedia.org/wiki/Hoeffding%27s_inequality

Gaussian Mixture Models

Higher order moments

Kurtosis

https://en.wikipedia.org/wiki/Kurtosis

blind source separation

ICA = Independent Component Analysis

https://en.wikipedia.org/wiki/Weibull_distribution

Maximum Likelihood estimation

Biased and unbiased estimators

Question to answer:

how confident are you about ... ?

focus on uncertainty ...

cauchy distribution – ratio of normals
https://en.wikipedia.org/wiki/Cauchy_distribution

beta distribution

bootstraping – estimating uncertainty of your estimator when CLT doesn't apply

negative binomial distribution - how many steps until N successes achieved

poisson approximation for rare events

hyper geometric distribution use cases – without replacement in finite population

More links:

Confidence Interval
- https://en.wikipedia.org/wiki/Confidence_interval

Cluster Analysis (finding groups (clusters) in data)
- https://en.wikipedia.org/wiki/Cluster_analysis

Student's t-distribution - estimating the mean for small sample sizes when **σ** is unknown.
- https://en.wikipedia.org/wiki/Student%27s_t-distribution

Chi-squared distribution - distribution of a sum of the squares of k independent standard normal random variables
- https://en.wikipedia.org/wiki/Chi-squared_distribution

F-distribution (also ANOVA - Analysis of Variance)
- https://en.wikipedia.org/wiki/F-distribution

Gamma distribution - a two-parameter family of continuous probability distributions.
Special cases – exponential and chi-squared distributions.
- https://en.wikipedia.org/wiki/Gamma_distribution

Convolution, Functional Analysis
– https://en.wikipedia.org/wiki/Convolution –
– https://en.wikipedia.org/wiki/Functional_analysis –

Extracting signal from Noise
– https://en.wikipedia.org/wiki/Signal-to-noise_ratio –
– https://en.wikipedia.org/wiki/Noise_reduction – (filtering, etc.). If we add N samples "s" of noisy data, then the signal will add proportionally to N, whereas noise will grow only as sqrt(N). So, if we add 100 samples, we will improve out Signal/Noise ratio ~10 times.

Monte Carlo Method - used to evaluate something by using randomly-generated samples.
The more samples – the more accurate the result.
- https://en.wikipedia.org/wiki/Monte_Carlo_method –

More links:

Stochastic Processes, Time Series Analysis
- https://en.wikipedia.org/wiki/Stochastic_process –
- https://en.wikipedia.org/wiki/Time_series –
- https://en.wikipedia.org/wiki/Random_walk –
- https://en.wikipedia.org/wiki/Brownian_motion –
- https://en.wikipedia.org/wiki/Diffusion ,
- https://en.wikipedia.org/wiki/Diffusion_process –
- https://en.wikipedia.org/wiki/Stationary_process –
- https://en.wikipedia.org/wiki/Stable_process –

Poisson Process
– https://en.wikipedia.org/wiki/Poisson_point_process –
– https://en.wikipedia.org/wiki/Dirac_delta_function –
– https://en.wikipedia.org/wiki/Compound_Poisson_process –

White Noise, Gaussian Noise, Brownian Noise
– https://en.wikipedia.org/wiki/White_noise –
– https://en.wikipedia.org/wiki/Gaussian_noise –
– https://en.wikipedia.org/wiki/Brownian_noise –

Markov Process
– https://en.wikipedia.org/wiki/Markov_chain –
– https://en.wikipedia.org/wiki/Markov_process –

Correlation function:
– https://en.wikipedia.org/wiki/Correlation_function –
– https://en.wikipedia.org/wiki/Autocorrelation –

Spectral Density, Fourier Analysis
– https://en.wikipedia.org/wiki/Spectral_density –
– https://en.wikipedia.org/wiki/Fourier_analysis –
– https://en.wikipedia.org/wiki/Fast_Fourier_transform –